Classification of Elements and Periodicity in Properties

Question1

First and second ionization enthalpies of lithium are $520 \mathrm{~kJ} \mathrm{~mol}^{-1}$ and $7297 \mathrm{~kJ} \mathrm{~mol}^{-1}$ respectively. Energy required to convert 3.5 mg lithium (g) into $\mathrm{Li}^{2+}(\mathrm{g})\left[\mathrm{Li}(\mathrm{g}) \rightarrow \mathrm{Li}^{2+}(\mathrm{g})\right]$ is ____ kJmol^{-1}. (nearest integer)

[Molar mass of $\mathrm{Li}=7 \mathrm{~g} \mathrm{~mol}^{-1}$]

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Answer: 4

Solution:

First, find the number of moles of lithium. We are given the mass of lithium as $3.5 \, \mathrm{mg} = 3.5 \times 10^{-3} \, \mathrm{g}$ and the molar mass as $7 \, \mathrm{g} \, \mathrm{mol}^{-1}$.

So,

Number of moles of $\mathrm{Li} = \frac{3.5 \times 10^{-3}}{7} = 5 \times 10^{-4}$

The process $\mathrm{Li}(\mathrm{g}) \rightarrow \mathrm{Li}^{2+}(\mathrm{g})$ involves removing two electrons. The energy required for this is equal to the sum of first and second ionization enthalpies of lithium.

Total ionization energy per mole of lithium:

$520 + 7297 = 7817 \, \mathrm{kJ} \, \mathrm{mol}^{-1}$

Now, calculate the energy for 5×10^{-4} moles of lithium:

Energy required $= 5 \times 10^{-4} \times 7817 = 3.9085 \, \mathrm{kJ}$

Therefore, the total energy required is approximately $4 \, \mathrm{kJ}$.

Question2

Which of the following represents the correct trend for the mentioned property?

A. F > P > S > B - First Ionization Energy

B. Cl > F > S > P - Electron Affinity

C. K > Al > Mg > B - Metallic character

D. $K_2O > Na_2O > MgO > Al_2O_3$ - Basic character

Choose the correct answer from the options given below :

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Options:

A.

B and C only

B.

A, B and D only

C.

A and B only

D.

A, B, C and D

Answer: B

Solution:

Ionisation Energy (Option A)

• Across a period (left ➜ right) atoms get smaller and the nucleus pulls more strongly, so the first ionisation energy (IE) goes up.
• Down a group (top ➜ bottom) atoms get bigger, so the IE goes down.

$F > P > S > B$ (IE order)

Metallic Character (Option C)

• Metallic character means how easily an atom loses electrons.
• Across a period it falls because atoms hold their electrons more tightly.
• Down a group it rises because outer electrons are farther from the nucleus.

$K > Mg > Al > B$ (Metallic character order)

Basic Nature of Oxides (Option D)

• Metal oxides are basic; non-metal oxides are acidic.
• As metallic character increases down a group, basic strength of the oxides also increases.

$K_2O > Na_2O > MgO > Al_2O_3$

Electron Affinity (Option B)

• Electron affinity (EA) usually becomes more negative from left to right because atoms want to gain electrons.
• Among the main-group elements, EA follows Group 17 > Group 16 > Group 15.

$Cl > F > S > P$

Question3

Given below are two statements :

Statement I : The correct order in terms of atomic/ionic radii is Al > Mg > Mg^{2+} > Al^{3+}.

Statement II : The correct order in terms of the magnitude of electron gain enthalpy is Cl > Br > S > O.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Statement I is true but **Statement II** is false

B.

Both **Statement I** and **Statement II** are true

C.

Statement I is false but **Statement II** is true

D.

Both **Statement I** and **Statement II** are false

Answer: C

Solution:

Statement I:

The order given is:

Al > Mg > Mg^{2+} > Al^{3+}

We are comparing *atomic and ionic radii.*

- **Mg (Z = 12)** → configuration: $[Ne]3s^2$
- **Al (Z = 13)** → configuration: $[Ne]3s^23p^1$

Across a period (from Mg to Al), the effective nuclear charge increases, so **atomic radius decreases**.

Therefore, **Mg > Al** (not Al > Mg).

For the ions:

- **Mg^{2+}** and **Al^{3+}** are isoelectronic (both have 10 electrons, same as Ne).

  Among isoelectronic species, the greater the nuclear charge, the smaller the radius.

  Hence, **$Al^{3+} < Mg^{2+}$**.

So the correct order should be:

$Mg > Al > Mg^{2+} > Al^{3+}$

Therefore, **Statement I is false.**

Statement II:

Electron gain enthalpy (EGE) refers to the energy change when an atom gains an electron.

The *more negative* the EGE, the greater the electron affinity.

The general trend for electron affinity (magnitude):

Cl > Br > S > O

Reasoning:

- Across a period and down a group, EGE usually becomes less negative (magnitude decreases down group 17).
- Halogens have high (negative) EGEs.
- Among main-group nonmetals, oxygen and sulfur have less negative EGE due to smaller size and electron–electron repulsion in the compact 2p orbital of oxygen.

Thus, **Cl > Br > S > O** is the correct order **in magnitude.**

So **Statement II is true.**

✅ **Final Answer:**

Option C: Statement I is false but Statement II is true.

Question4

Given below are two statements :

Statement I : $\mathrm{C}<\mathrm{O}<\mathrm{N}<\mathrm{F}$ is the correct order in terms of first ionization enthalpy values.

Statement II : $\mathrm{S}>\mathrm{Se}>\mathrm{Te}>\mathrm{Po}>\mathrm{O}$ is the correct order in terms of the magnitude of electron gain enthalpy values.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are false

B.

Statement I is false but Statement II is true

C.

Statement I is true but Statement II is false

D.

Both Statement I and Statement II are true

Answer: D

Solution:

Statement I (First ionization enthalpy)

Across a period, first ionization enthalpy generally **increases** due to increasing nuclear charge and decreasing atomic size.

But there is an important NCERT exception:

- **Nitrogen (N)** has a half-filled configuration $2p^3$, which is extra stable.
- **Oxygen (O)** has configuration $2p^4$, where one p-orbital contains a pair of electrons, causing extra electron–electron repulsion, so it is easier to remove one electron.

So,

$\mathrm{IE}_1(\mathrm{O}) < \mathrm{IE}_1(\mathrm{N})$

Also, in general:

$\mathrm{C} < \mathrm{O} < \mathrm{N} < \mathrm{F}$

✅ **Statement I is true.**

Statement II (Magnitude of electron gain enthalpy)

Electron gain enthalpy usually becomes **less negative** down a group due to increasing size (incoming electron feels less attraction).

But for group 16, NCERT notes:

- Oxygen is very small, so the incoming electron experiences strong repulsion in the compact $2p$ subshell.
- Hence oxygen has **less negative** electron gain enthalpy than sulphur.
- Sulphur has the **most negative** (highest magnitude) electron gain enthalpy in this group.

Therefore, the **magnitude** order is:

$\mathrm{S} > \mathrm{Se} > \mathrm{Te} > \mathrm{Po} > \mathrm{O}$

✅ **Statement II is true.**

Correct option: D

✅ **Both Statement I and Statement II are true.**

Question5

The correct trend in the first ionization enthalpies of the elements in the 3^{rd} period of periodic table is :

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Options:

A.

$\mathrm{Si} < \mathrm{S} < \mathrm{Al} < \mathrm{P} < \mathrm{Cl}$

B.

$\mathrm{Al} < \mathrm{Si} < \mathrm{S} < \mathrm{P} < \mathrm{Cl}$

C.

$\mathrm{S} < \mathrm{Si} < \mathrm{Al} < \mathrm{P} < \mathrm{Cl}$

D.

$\mathrm{Al} < \mathrm{S} < \mathrm{P} < \mathrm{Si} < \mathrm{Cl}$

Answer: B

Solution:

In general on moving from left to right in a period ionization energy increases as $\mathbf{Z}_{\text{eff}}$ increases.

$\mathrm{Al} < \mathrm{Si} < \mathrm{S} < \mathrm{P} < \mathrm{Cl}$

(Ionisation energy of phosphorus is more because of half filled stable configuration)

Question6

Elements X and Y belong to Group 15. The difference between the electronegativity values of ' X ' and phosphorus is higher than that of the difference between phosphorus and ' Y '. ' X ' & ' Y ' are respectively

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Options:

A.

As & Bi

B.

Bi & N

C.

N & As

D.

As & Sb

Answer: C

Solution:

As per the NCERT textbook, electronegativity is the ability of an atom to attract a shared pair of electrons. For Group 15, the electronegativity **decreases as we go down the group**.

Let's look at their approximate electronegativity values (on the Pauling scale):

- $EN(N) = 3.0$
- $EN(P) = 2.1$
- $EN(As) = 2.0$
- $EN(Sb) = 1.9$
- $EN(Bi) = 1.9$

Notice a very important point: The drop in electronegativity from Nitrogen (N) to Phosphorus (P) is very large ($3.0 - 2.1 = 0.9$). However, for the elements below Phosphorus, the values decrease very slowly.

The question states: "The difference between the electronegativity values of 'X' and phosphorus is higher than that of the difference between phosphorus and 'Y'."

Let's write this mathematically. Let EN stand for electronegativity. The absolute difference is what matters.

$|EN(X) - EN(P)| > |EN(P) - EN(Y)|$

This means we are looking for an element 'X' whose electronegativity is very different from Phosphorus, and an element 'Y' whose electronegativity is very close to Phosphorus.

Now, let's check each option using the electronegativity values.

- **Option A: X = As & Y = Bi**
  - Difference 1: $|EN(X) - EN(P)| = |EN(As) - EN(P)| = |2.0 - 2.1| = 0.1$
  - Difference 2: $|EN(P) - EN(Y)| = |EN(P) - EN(Bi)| = |2.1 - 1.9| = 0.2$
  - Is $0.1 > 0.2$? No. So, this option is incorrect.
- **Option B: X = Bi & Y = N**
  - Difference 1: $|EN(X) - EN(P)| = |EN(Bi) - EN(P)| = |1.9 - 2.1| = 0.2$
  - Difference 2: $|EN(P) - EN(Y)| = |EN(P) - EN(N)| = |2.1 - 3.0| = 0.9$
  - Is $0.2 > 0.9$? No. So, this option is incorrect.
- **Option C: X = N & Y = As**
  - Difference 1: $|EN(X) - EN(P)| = |EN(N) - EN(P)| = |3.0 - 2.1| = 0.9$

- Difference 2: $|EN(P) - EN(Y)| = |EN(P) - EN(As)| = |2.1 - 2.0| = 0.1$
- Is $0.9 > 0.1$? Yes. This matches the condition given in the question.

- **Option D: X = As & Y = Sb**
  - Difference 1: $|EN(X) - EN(P)| = |EN(As) - EN(P)| = |2.0 - 2.1| = 0.1$
  - Difference 2: $|EN(P) - EN(Y)| = |EN(P) - EN(Sb)| = |2.1 - 1.9| = 0.2$
  - Is $0.1 > 0.2$? No. So, this option is incorrect.

Therefore, X is Nitrogen (N) and Y is Arsenic (As).

Question7

Given below are two statements :

Statement I : The second ionisation enthalpy of Na is larger than the corresponding ionisation enthalpy of Mg .

Statement II : The ionic radius of O^{2-} is larger than that of F^{-}.

In the light of the above statements, choose the *correct* answer from the options given below :

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Options:

A.

Statement I is true but Statement II is false

B.

Statement I is false but Statement II is true

C.

Both Statement I and Statement II are true

D.

Both Statement I and Statement II are false

Answer: C

Solution:

Statement I:

Na: $1s^2\,2s^2\,2p^6\,3s^1$

After first ionisation, Na^+ becomes $1s^2\,2s^2\,2p^6$ (noble gas configuration).

So, the **second ionisation enthalpy** of Na means removing an electron from a **stable noble gas core**, which needs **very high energy**.

Mg: $1s^2\,2s^2\,2p^6\,3s^2$

After first ionisation, Mg^+ becomes $1s^2\,2s^2\,2p^6\,3s^1$.

So, the **second ionisation enthalpy** of Mg removes the remaining **valence $3s$ electron**, which needs **less energy** than removing from a noble gas core.

Hence, $IE_2(\text{Na}) > IE_2(\text{Mg})$.

So, **Statement I is true**.

Statement II:

O^{2-} has $8 + 2 = 10$ electrons, and F^- has $9 + 1 = 10$ electrons.

So, O^{2-} and F^- are **isoelectronic** (same number of electrons).

In an **isoelectronic series**, ionic radius **decreases** as **nuclear charge (Z) increases**, because higher Z pulls electrons more strongly.

Here, $Z(\text{O}) = 8$ and $Z(\text{F}) = 9$.

So, F^- (higher Z) will be **smaller** than O^{2-}.

Therefore, $r(O^{2-}) > r(F^-)$.

So, **Statement II is true**.

Correct option: Option C

Both Statement I and Statement II are true.

Question8

Given below are two statements :

Statement I : $\text{K} > \text{Mg} > \text{Al} > \text{B}$ is the correct order in terms of metallic character.

Statement II : Atomic radius is always greater than the ionic radius for any element.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Both Statement I and Statement II are false

B.

Statement I is false but Statement II is true

C.

Statement I is true but Statement II is false

D.

Both Statement I and Statement II are true

Answer: C

Solution:

Metallic character means the tendency of an element to lose electrons and form a positive ion. In general (as per NCERT trends), metallic character is more in **s-block** elements than in **p-block** elements.

So, elements like K and Mg (s-block) show more metallic character than Al and B (p-block).

Now, compare atomic radius and ionic radius: if an atom forms a **cation**, its ionic radius becomes smaller than the atomic radius. If an atom forms an **anion**, its ionic radius becomes larger than the atomic radius.

Therefore, **ionic radius is not always smaller** than atomic radius. Specifically, anionic radius is greater than atomic radius, but cationic radius is always less than atomic radius for any element.

Question9

The correct order of $\mathrm{C}, \mathrm{N}, \mathrm{O}$ and F in terms of second ionisation potential is

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Options:

A.

$\mathrm{C}<\mathrm{N}<\mathrm{F}<\mathrm{O}$

B.

$\mathrm{C}<\mathrm{F}<\mathrm{N}<\mathrm{O}$

C.

$\mathrm{F}<\mathrm{N}<\mathrm{C}<\mathrm{O}$

D.

$\mathrm{C}<\mathrm{O}<\mathrm{N}<\mathrm{F}$

Answer: A

Solution:

To compare second ionization potential configuration of mono-cation is observed

C^{+}	N^{+}	O^{+}	F^{+}
[He] $2s^2sp^1$	[He] $2s^22p^2$	[He]$2s^22p^3$ Half-filled stable.	[He]$2s^22p^4$

2^{nd} IE order
$O > F > N > C$

Question10

In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively.

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Options:

A.

Rb & Br

B.

Na & Cl

C.

K&Br

D.

K&Se

Answer: C

Solution:

In **period 4** (from K to Kr), the **atomic radius decreases** as we move from left to right because:

- **Nuclear charge increases** (Z increases),
- Electrons are added in the **same principal shell** ($n = 4$),
- So the **effective nuclear charge** on valence electrons increases and pulls them closer.

Hence:

- The **largest atomic radius** is for the **first element** of the period: $\boxed{K}$.
- The **smallest atomic radius** (as per NCERT trend, minimum at halogen) is for the **halogen** in that period: $\boxed{Br}$.

So, the elements with **highest and lowest atomic radii** respectively are:

$\boxed{\text{K \& Br}}$

Correct option: C

Question11

Consider the elements N, P, O, S, Cl and F. The number of valence electrons present in the elements with most and least metallic character from the above list is respectively.

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Options:

A.

7 and 5

B.

6 and 7

C.

5 and 7

D.

5 and 6

Answer: C

Solution:

Metallic character **increases down a group** and **decreases from left to right** across a period (NCERT trend).

Given elements:

- Period 2: $N,\ O,\ F$
- Period 3: $P,\ S,\ Cl$

1) Most metallic element

In **period 3**, metallic character is higher than in period 2, and within period 3 it decreases left to right:

$P > S > Cl$

So, P is the **most metallic** among the given elements.

Valence electrons in P (Group 15) $= 5$.

2) Least metallic element

Least metallic means **most non-metallic**, which is highest towards the **top-right** of the periodic table. Among the given, F is the most non-metallic.

Valence electrons in F (Group 17) $= 7$.

Final Answer

Number of valence electrons (most metallic, least metallic) $= (5,\ 7)$.

Option C: 5 and 7

Question12

Given below are two statements :

Statement (I) : The first ionisation enthalpy of the elements Na, Mg, Cl and Ar follows the order Na > Mg > Cl > Ar.

Statement (II) : Among Ca, Al, Fe and B, the third ionisation enthalpy is very high for Ca.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

Both **Statement I** and **Statement II** are true

B.

Both **Statement I** and **Statement II** are false

C.

Statement I is true but **Statement II** is false

D.

Statement I is false but **Statement II** is true

Answer: D

Solution:

For **Statement (I)**, let us check the trend of **first ionisation enthalpy**.

Across a period, first ionisation enthalpy generally **increases** from left to right because effective nuclear charge increases.

For the elements given:

- Na
- Mg
- Cl
- Ar

the correct order of first ionisation enthalpy is

$\mathrm{Na} < \mathrm{Mg} < \mathrm{Cl} < \mathrm{Ar}$

But the statement says

$\mathrm{Na} > \mathrm{Mg} > \mathrm{Cl} > \mathrm{Ar}$

This is clearly **wrong**.

So, **Statement (I) is false**.

For **Statement (II)**, we compare the **third ionisation enthalpy** of Ca, Al, Fe and B.

Third ionisation enthalpy means removal of the third electron, that is, from the species M^{2+}.

Now consider calcium:

- Ca has electronic configuration

  $[\mathrm{Ar}]\, 4s^2$

- After losing two electrons,

  $\mathrm{Ca}^{2+} = [\mathrm{Ar}]$

  which is a **noble gas configuration**.

Removing the third electron from Ca^{2+} means breaking a very stable noble gas configuration, so the **third ionisation enthalpy is very high**.

Hence, **Statement (II) is true**.

Therefore:

- **Statement (I) is false**
- **Statement (II) is true**

So, the correct answer is:

$\boxed{\text{Option D}}$

Question13

The correct set that contains all kinds (basic, acidic, amphoteric and neutral) of oxides is :

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Options:

A.

Na_2O, K_2O, Al_2O_3 and As_2O_3

B.

Al_2O_3, As_2O_3, CO and NO

C.

K_2O, Cl_2O_7, As_2O_3 and NO

D.

Na_2O, N_2O, Al_2O_3 and CO

Answer: C

Solution:

To find the correct set, we need one oxide of each type:

- **Basic oxide**
- **Acidic oxide**
- **Amphoteric oxide**
- **Neutral oxide**

Now let us check the given oxides.

Important classifications

- **Basic oxides**: Usually metallic oxides like Na_2O, K_2O
- **Acidic oxides**: Usually non-metallic oxides like Cl_2O_7
- **Amphoteric oxides**: Oxides that react with both acids and bases, like Al_2O_3, As_2O_3
- **Neutral oxides**: Neither acidic nor basic, like CO, NO, N_2O

Option A

$Na_2O,\ K_2O,\ Al_2O_3,\ As_2O_3$

- Na_2O = basic
- K_2O = basic
- Al_2O_3 = amphoteric
- As_2O_3 = amphoteric

This set does **not** contain acidic and neutral oxides.

So, **Option A is incorrect**.

Option B

$Al_2O_3,\ As_2O_3,\ CO,\ NO$

- Al_2O_3 = amphoteric
- As_2O_3 = amphoteric
- CO = neutral

- NO = neutral

This set does **not** contain basic and acidic oxides.

So, **Option B is incorrect**.

Option C

$K_2O,\ Cl_2O_7,\ As_2O_3,\ NO$

- K_2O = basic
- Cl_2O_7 = acidic
- As_2O_3 = amphoteric
- NO = neutral

This set contains **all four kinds** of oxides.

So, **Option C is correct**.

Option D

$Na_2O,\ N_2O,\ Al_2O_3,\ CO$

- Na_2O = basic
- N_2O = neutral
- Al_2O_3 = amphoteric
- CO = neutral

This set does **not** contain any acidic oxide.

So, **Option D is incorrect**.

Final Answer

$\boxed{\text{Option C}}$

Question14

The 1^{st} ionization enthalpy for Mg is $+737\ \text{kJ/mol}$. The most probable estimated value of the 2^{nd} ionization enthalpy of Mg is ____ .

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Options:

A.

$-906\ \text{kJ/mol}$

B.

$-856\ \text{kJ/mol}$

C.

$+1450\ \mathrm{kJ/mol}$

D.

$+590\ \mathrm{kJ/mol}$

Answer: C

Solution:

For magnesium, the electronic configuration is

$\mathrm{Mg} = 1s^2\, 2s^2\, 2p^6\, 3s^2$

The first ionization enthalpy corresponds to:

$\mathrm{Mg(g)} \rightarrow \mathrm{Mg^+(g)} + e^-$

This is given as

$\Delta_i H_1 = +737\,\mathrm{kJ\ mol^{-1}}$

Now the second ionization enthalpy corresponds to removing one more electron from $\mathrm{Mg^+}$:

$\mathrm{Mg^+(g)} \rightarrow \mathrm{Mg^{2+}(g)} + e^-$

After losing one electron, $\mathrm{Mg^+}$ has configuration

$1s^2\, 2s^2\, 2p^6\, 3s^1$

The second electron is still being removed from the same outermost shell, but now from a positively charged ion. So this electron is held more strongly than in neutral Mg.

Hence:

- second ionization enthalpy must be greater than first ionization enthalpy,
- ionization enthalpy is always positive.

So options with negative values are impossible, and $+590\,\mathrm{kJ\ mol^{-1}}$ is less than the first ionization enthalpy, so it is also not possible.

Therefore, the most probable value is

$\boxed{+1450\,\mathrm{kJ\ mol^{-1}}}$

So, the correct option is C.

Question15

Given below are two statements:

Statement I: The correct order of electronegativity of fluorine, oxygen and nitrogen is $\mathrm{F > O > N}$.

Statement II: The oxidation state of oxygen in $\mathrm{OF_2}$ is +2 and in $\mathrm{Na_2O}$ is -2 .

In the light of the above statements, choose the correct answer from the options given below

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: A

Solution:

Statement I:

Electronegativity generally increases across a period from left to right.

For the elements nitrogen, oxygen, and fluorine in the second period, the correct order is

$\mathrm{F} > \mathrm{O} > \mathrm{N}$

So, Statement I is true.

Statement II:

Let us check the oxidation state of oxygen in both compounds.

1. In OF_2

Fluorine is the most electronegative element, so its oxidation state is always -1.

Let the oxidation state of oxygen be x.

Then,

$x + 2(-1) = 0$

$x - 2 = 0$

$x = +2$

So, in OF_2, oxygen has oxidation state $+2$.

2. In $\mathrm{Na}_2\mathrm{O}$

Sodium has oxidation state $+1$.

Let the oxidation state of oxygen be x.

Then,

$2(+1) + x = 0$

$2 + x = 0$

$x = -2$

So, in Na_2O, oxygen has oxidation state -2.

Hence, Statement II is also true.

Therefore, the correct answer is:

$\boxed{\text{Option A}}$

Both Statement I and Statement II are true.

Question16

Given below are two statements :

Statement I : The number of pairs among $[Al_2O_3, Cr_2O_3]$, $[Cl_2O_7, Mn_2O_7]$, $[Na_2O, V_2O_3]$ and $[CO, N_2O]$ that contain oxides of same nature (acidic, basic, neutral or amphoteric) is 4 .

Statement II : Among Na_2O, Al_2O_3, CO and Cl_2O_7, the most basic and acidic oxides are Na_2O and Cl_2O_7, respectively.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: A

Solution:

Evaluating Statement I:

We need to determine the chemical nature of each pair of oxides:

1. Pair 1: $[Al_2O_3, Cr_2O_3]$

- Al_2O_3 (Aluminium oxide) is a standard example of an amphoteric oxide as it reacts with both acids and bases.
- Cr_2O_3 (Chromium(III) oxide) has chromium in the $+3$ oxidation state. Transition metal oxides in intermediate oxidation states are generally amphoteric.

Both oxides are amphoteric, so they have the same nature.

2. Pair 2: $[Cl_2O_7, Mn_2O_7]$

- Cl_2O_7 is a non-metal oxide in a very high oxidation state $(+7)$, which makes it strongly acidic.
- Mn_2O_7 contains manganese in its highest oxidation state $(+7)$. Transition metals in their highest oxidation states form highly acidic oxides.

Both oxides are acidic, so they have the same nature.

3. Pair 3: $[Na_2O, V_2O_3]$

- Na_2O is an alkali metal oxide and is strongly basic.
- V_2O_3 contains vanadium in a lower oxidation state $(+3)$. Lower oxidation state oxides of d-block elements are typically basic in nature.

Both oxides are basic, so they have the same nature.

4. Pair 4: $[CO, N_2O]$

- Both CO (carbon monoxide) and N_2O (nitrous oxide) are classic examples of non-metal oxides that are completely neutral.

Both oxides are neutral, so they have the same nature.

Since all 4 pairs contain oxides of the same chemical nature, Statement I is true.

Evaluating Statement II:

We are given four oxides: Na_2O, Al_2O_3, CO, and Cl_2O_7. Let us list their nature:

- Na_2O is a basic oxide.
- Al_2O_3 is an amphoteric oxide.
- CO is a neutral oxide.
- Cl_2O_7 is an acidic oxide.

From this list, it is clear that Na_2O is the most basic because it is an oxide of a highly electropositive Group 1 metal. Similarly, Cl_2O_7 is the most acidic because it is an oxide of a highly electronegative halogen in its highest oxidation state. Thus, Statement II is true.

Conclusion:

Since both Statement I and Statement II are factually correct, the correct answer is Option A.

Option A

Question17

In a period, the first ionisation enthalpy of the element at extreme left and the negative electron gain enthalpy of the extreme right element, except noble gases, are respectively.

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Options:

A.

lowest and lowest

B.

highest and lowest

C.

lowest and highest

D.

highest and highest

Answer: C

Solution:

The element on the far left of a period belongs to Group 1. These elements have only one electron in their outermost shell, so they lose this electron easily. Therefore, they have the lowest first ionisation enthalpy (IE).

The element on the far right of a period, except noble gases, belongs to Group 17 (halogens). These atoms need just one electron to complete their outermost shell, so they strongly attract an extra electron. Hence, they have the most negative value of electron gain enthalpy (ΔH_{eg}).

Question18

Match List - I with List - II.

List - I Electronic configuration of neutral atom (where $n=2$)		List - II 1^{st} Ionization Energy ($\mathbf{kJ\ mol^{-1}}$)	
A.	ns^2	I.	2080
B.	$\mathrm{ns}^2\mathrm{np}^1$	II.	899
C.	ns^2np^3	III.	800
D.	ns^2np^6	IV.	1402

Choose the correct answer from the options given below :

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Options:

A.

A-II, B-III, C-IV, D-I

B.

A-IV, B-III, C-II, D-I

C.

A-III, B-II, C-IV, D-I

D.

A-III, B-II, C-I, D-IV

Answer: A

Solution:

Ionization energy is the energy needed to remove the most loosely held (outermost) electron from a neutral atom.

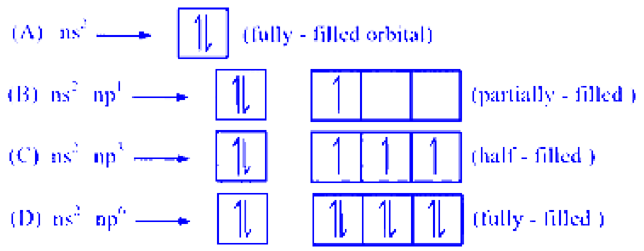
Ionization energy generally increases when the effective nuclear charge increases. This happens because the nucleus attracts the valence electron more strongly, so more energy is required to remove it.

Also, some electronic configurations are extra stable. Electrons in stable configurations are harder to remove, so their ionization energy is higher.

Fully filled orbitals have greater ionisation energy due to stable electronic configuration.

Order of I.E:

$\text{fully-filled orbital} > \text{Half-filled orbital} > \text{Partially filled orbital}$



Here $n = 2$, so we are comparing configurations of elements in the 2nd period. As we move from s-block to p-block in the same period, atomic size decreases and effective nuclear charge increases, so ionisation energy generally increases.

Using stability ideas: the configuration ns^2np^6 (noble gas, fully filled) has the maximum ionisation energy, and ns^2np^1 has comparatively lower ionisation energy.

So the decreasing order is: $D > C > A > B$

Ans. (A)

Question19

Which of the following electronegativity order is incorrect?

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Options:

A. $\mathrm{S} < \mathrm{Cl} < \mathrm{O} < \mathrm{F}$

B. $\mathrm{Al} < \mathrm{Si} < \mathrm{C} < \mathrm{N}$

C. $\mathrm{Al} < \mathrm{Mg} < \mathrm{B} < \mathrm{N}$

D. $\mathrm{Mg} < \mathrm{Be} < \mathrm{B} < \mathrm{N}$

Answer: C

Solution:

Let’s compare the known electronegativities (Pauling scale) of the elements in each option:

F ~ 3.98

O ~ 3.44

N ~ 3.04

Cl ~ 3.16

C ~ 2.55

B ~ 2.04

S ~ 2.58 (sometimes listed as 2.5–2.58)

Si ~ 1.90

Al ~ 1.61

Be ~ 1.57

Mg ~ 1.31

Now, check each statement:

Option A: $\mathrm{S} < \mathrm{Cl} < \mathrm{O} < \mathrm{F}$

S (2.58) < Cl (3.16) < O (3.44) < F (3.98)

This order is **correct**.

Option B: $\mathrm{Al} < \mathrm{Si} < \mathrm{C} < \mathrm{N}$

Al (1.61) < Si (1.90) < C (2.55) < N (3.04)

This order is **correct**.

Option C: $\mathrm{Al} < \mathrm{Mg} < \mathrm{B} < \mathrm{N}$

Actual electronegativities are Al (1.61) and Mg (1.31).

The statement says “Al < Mg,” which would mean 1.61 < 1.31, which is **wrong**.

Therefore, **Option C** is **incorrect**.

Option D: $\mathrm{Mg} < \mathrm{Be} < \mathrm{B} < \mathrm{N}$

Mg (1.31) < Be (1.57) < B (2.04) < N (3.04)

This order is **correct**.

Answer: Option C is the incorrect order.

Question20

Match List-I with List-II.

	List - I		List - II
(A)	$\mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{F}^{-}$	(I)	Ionisation Enthalpy
(B)	$\mathrm{B} < \mathrm{C} < \mathrm{O} < \mathrm{N}$	(II)	Metallic character
(C)	$\mathrm{B} < \mathrm{Al} < \mathrm{Mg} < \mathrm{K}$	(III)	Electronegativity
(D)	$\mathrm{Si} < \mathrm{P} < \mathrm{S} < \mathrm{Cl}$	(IV)	Ionic radii

Choose the correct answer from the options given below :

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Options:

A. (A)-(IV), (B)-(I), (C)- (II), (D)-(III)

B. (A)-(III), (B)-(IV), (C)- (II), (D)-(I)

C. (A)-(II), (B)-(III), (C)- (IV), (D)-(I)

D. (A)-(IV), (B)-(I), (C)- (III), (D)-(II)

Answer: A

Solution:

(A) $\mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{F}^{-}$

All four ions ($\mathrm{Al}^{3+}, \mathrm{Mg}^{2+}, \mathrm{Na}^{+}, \mathrm{F}^{-}$) are **isoelectronic** (each has 10 electrons). For isoelectronic species, ionic radii **decrease** as the positive nuclear charge **increases**, which is why the smallest ion here is Al^{3+} (Z=13) and the largest is F^{-} (Z=9). Thus, this ordering is one of **increasing ionic radius**.

$\boxed{(A) \longrightarrow \text{(IV) Ionic radii}}$

(B) $\mathrm{B} < \mathrm{C} < \mathrm{O} < \mathrm{N}$

Check **first ionization enthalpies** (IE_1):

B: ≈ 801\,kJ/mol

C: ≈ 1086\,kJ/mol

O: ≈ 1314\,kJ/mol

N: ≈ 1402\,kJ/mol

Hence, the order of increasing IE_1 is

$B < C < O < N.$

This matches the given sequence exactly.

$\boxed{(B) \longrightarrow \text{(I) Ionisation enthalpy}}$

(C) $\mathrm{B} < \mathrm{Al} < \mathrm{Mg} < \mathrm{K}$

Consider **metallic character** (the tendency to lose electrons easily, show metallic properties). Across a period (left to right), metallic character **decreases**; down a group, it **increases**.

B (metalloid) has the least metallic character here.

Al (group 13 metal) is more metallic than B.

Mg (group 2 metal) is typically more metallic than Al.

K (group 1 metal) is the most metallic among these.

Thus, $\mathrm{B} < \mathrm{Al} < \mathrm{Mg} < \mathrm{K}$ is an order of **increasing metallic character**.

$\boxed{(C) \longrightarrow \text{(II) Metallic character}}$

(D) $\mathrm{Si} < \mathrm{P} < \mathrm{S} < \mathrm{Cl}$

Check **electronegativities**:

Si: ≈ 1.90

P: ≈ 2.19

S: ≈ 2.58

Cl: ≈ 3.16

They increase in the order

Si < P < S < Cl,

which matches the given sequence for **increasing electronegativity**.

$\boxed{(D) \longrightarrow \text{(III) Electronegativity}}$

Final Matching

$(A) \to (IV), \quad (B) \to (I), \quad (C) \to (II), \quad (D) \to (III).$

Looking at the choices given:

Option A: $(A) - (IV), (B) - (I), (C) - (II), (D) - (III)$

This is exactly what we found.

Answer: Option A

Question21

Given below are two statements :

Statement (I) : An element in the extreme left of the periodic table forms acidic oxides.

Statement (II) : Acid is formed during the reaction between water and oxide of a reactive element present in the extreme right of the periodic table.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A. Both Statement I and Statement II are false

B. Both Statement I and Statement II are true

C. Statement I is true but Statement II is false

D. Statement I is false but Statement II is true

Answer: D

Solution:

First, let us re-state the two statements clearly:

Statement (I): *An element in the extreme left of the periodic table forms acidic oxides.*

Statement (II): *Acid is formed during the reaction between water and oxide of a reactive element present in the extreme right of the periodic table.*

Analyzing Statement (I)

The *extreme left* of the periodic table corresponds to the alkali metals (Group 1) and alkaline earth metals (Group 2).

These metals typically form *basic* (or occasionally amphoteric, in the case of some Group 2 elements) oxides, **not acidic** oxides.

For example, $\mathbf{Na_2O}$, $\mathbf{K_2O}$, **MgO**, **CaO**, etc., all form *basic* solutions (e.g., $Na_2O + H_2O \rightarrow 2\ NaOH$).

Hence, **Statement (I)**—that an element in the extreme left forms acidic oxides—is **false**.

Analyzing Statement (II)

The *extreme right* of the periodic table corresponds to the nonmetals in Groups 15, 16, 17 (and noble gases in Group 18).

Nonmetal oxides (such as those of sulfur, phosphorus, chlorine) are generally *acidic*.

When these oxides dissolve in water, they typically form acids.

Example: $SO_3 + H_2O \rightarrow H_2SO_4$ (sulfuric acid)

Example: $P_2O_5 + 3\ H_2O \rightarrow 2\ H_3PO_4$ (phosphoric acid)

Example: $Cl_2O_7 + H_2O \rightarrow 2\ HClO_4$ (perchloric acid)

Thus, **Statement (II)**—that acid is formed when water reacts with an oxide of a reactive element in the extreme right—is **true**.

Conclusion

Statement (I) is **false**.

Statement (II) is **true**.

Therefore, the correct choice is:

> **(D) Statement (I) is false but Statement (II) is true.**

Question22

The element that does not belong to the same period of the remaining elements (modern periodic table) is:

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Options:

A. Platinum

B. Osmium

C. Iridium

D. Palladium

Answer: D

Solution:

Let's analyze the periods for each element:

Platinum (Pt) has an atomic number of 78, which places it in period 6.

Osmium (Os) has an atomic number of 76, also in period 6.

Iridium (Ir) has an atomic number of 77, which is in period 6 as well.

Palladium (Pd) has an atomic number of 46, placing it in period 5.

Since Palladium is in period 5 while the other elements are in period 6, the element that does not belong to the same period is:

Option D: Palladium.

Question23

Given below are the atomic numbers of some group 14 elements. The atomic number of the element with lowest melting point is :

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Options:

A. 14

B. 50

C. 6

D. 82

Answer: B

Solution:

Order of M.P. of group 14: C > Si > Ge > Pb > Sn

Element	M.P. ($^{\circ}$C)
Z = 6 = C	3730
Z = 14 = Si	1410
Z = 32 = Ge	937
Z = 50 = Sn	232
Z = 82 = Pb	327

Question24

Which of the following statements are NOT true about the periodic table?

A. The properties of elements are function of atomic weights.

B. The properties of elements are function of atomic numbers.

C. Elements having similar outer electronic configurations are arranged in same period.

D. An element's location reflects the quantum numbers of the last filled orbital.

E. The number of elements in a period is same as the number of atomic orbitals available in energy level that is being filled.

Choose the correct answer from the options given below:

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Options:

A. A, C and E Only

B. A and E Only

C. B, C and E Only

D. D and E Only

Answer: A

Solution:

Let's analyze each statement one by one:

A. The properties of elements are function of atomic weights.

Originally, elements were arranged by atomic weights, but it was later understood that their properties depend on the atomic number (the number of protons) rather than the atomic weight.

Thus, this statement is NOT true.

B. The properties of elements are function of atomic numbers.

The periodic table is arranged by increasing atomic number, and an element's properties directly relate to its number of protons and the corresponding electron configuration.

This statement is true.

C. Elements having similar outer electronic configurations are arranged in same period.

Elements that have similar outer electronic configurations are actually grouped in the same column (group) rather than the same row (period).

Therefore, this statement is NOT true.

D. An element's location reflects the quantum numbers of the last filled orbital.

The arrangement of elements corresponds to the electron configurations determined by quantum numbers, especially for the last electron added.

This statement is true.

E. The number of elements in a period is same as the number of atomic orbitals available in energy level that is being filled.

In an energy level, the number of atomic orbitals does not directly equal the number of elements in a period. For example, period 2 has 1s filled, and then the 2s and 2p orbitals are filled (a total of 4 orbitals), yet there are 8 elements because each orbital can accommodate 2 electrons.

Hence, this statement is NOT true.

So, the statements that are NOT true about the periodic table are A, C, and E.

Thus, the correct answer is:

Option A: A, C and E Only.

Question25

Given below are two statements :

Statement (I) : The first ionization energy of Pb is greater than that of Sn .

Statement (II) : The first ionization energy of Ge is greater than that of Si .

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A. Statement I is false but Statement II is true

B. Both Statement I and Statement II are false

C. Both Statement I and Statement II are true

D. Statement I is true but Statement II is false

Answer: D

Solution:

Order of I.E. (in KJ/mol) :

$$\begin{array}{ccccccccc} C & > & Si & > & Ge & > & Sn & > & Pb \\ 1086 & & 786 & & 761 & & 708 & & 715 \end{array}$$

Question26

The successive 5 ionisation energies of an element are $800, 2427, 3658, 25024$ and $32824 \mathrm{~kJ/mol}$, respectively. By using the above values predict the group in which the above element is present :

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Options:

A. Group 4

B. Group 2

C. Group 13

D. Group 14

Answer: C

Solution:

We can determine the number of valence electrons by looking for a large jump in successive ionization energies. Here's how we can analyze the data:

The ionization energies given are:

$IE_1 = 800\,\text{kJ/mol}$

$IE_2 = 2427\,\text{kJ/mol}$

$IE_3 = 3658\,\text{kJ/mol}$

$IE_4 = 25024\,\text{kJ/mol}$

$IE_5 = 32824\,\text{kJ/mol}$

Notice that:

The first three ionization energies are relatively low.

There is a significant jump between the third and fourth ionization energy.

What does the gap indicate?

The large jump from IE_3 to IE_4 means that after removing the first three electrons (which are the valence electrons), the next electron to be removed (a core electron) is much more strongly bound.

This implies that the element has 3 valence electrons.

Which group does an element with 3 valence electrons belong to?

Elements with 3 valence electrons are found in Group 13 of the periodic table.

Thus, the element most likely belongs to Group 13.

Correct Answer: Option C (Group 13).

Question27

Consider the following elements $\mathrm{In}, \mathrm{Tl}, \mathrm{Al}, \mathrm{Pb}, \mathrm{Sn}$ and Ge .The most stable oxidation states of elements with highest and lowest first ionisation enthalpies, respectively, are

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Options:

A. +4 and +1

B. +4 and +3

C. +1 and +4

D. +2 and +3

Answer: B

Solution:

Among Al, In, Tl, Ge, Sn, Pb, the metal having highest IE_1 is Ge and lowest IE_1 is In.

Most stable oxidation state of Ge is +4 and In is +3 .

Question28

The incorrect decreasing order of atomic radii is

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Options:

A. $\mathrm{Si} > \mathrm{P} > \mathrm{Cl} > \mathrm{F}$

B. $\mathrm{Mg} > \mathrm{Al} > \mathrm{C} > \mathrm{O}$

C. $\mathrm{Al} > \mathrm{B} > \mathrm{N} > \mathrm{F}$

D. $\mathrm{Be} > \mathrm{Mg} > \mathrm{Al} > \mathrm{Si}$

Answer: D

Solution:

Let's analyze the options based on the periodic trends in atomic radii. Recall two key trends:

Atomic radii decrease as you move from left to right in a period due to increasing effective nuclear charge.

Atomic radii increase as you move down a group because additional electron shells are added.

We'll review each option:

Option A: $\mathrm{Si} > \mathrm{P} > \mathrm{Cl} > \mathrm{F}$

Silicon (Si), phosphorus (P), and chlorine (Cl) are in the same period, with a gradual decrease in atomic radius from left to right.

Fluorine (F) is in an earlier period (it has only 2 electron shells) and is much smaller.

Hence, the order follows the expected trend.

Option B: $\mathrm{Mg} > \mathrm{Al} > \mathrm{C} > \mathrm{O}$

Magnesium (Mg) and aluminum (Al) are period 3 elements, whereas carbon (C) and oxygen (O) belong to period 2.

Period 3 elements generally have larger radii than period 2 counterparts.

The order is consistent: Mg and Al are larger than C and O, and within each period, radii decrease from left to right.

Option C: Al $>$ B $>$ N $>$ F

Aluminum (Al) is in period 3 and is expected to be larger than boron (B), nitrogen (N), and fluorine (F) from period 2.

Within period 2 (B, N, F), the atomic radii decrease from left to right.

Thus, this order is correct as well.

Option D: Be $>$ Mg $>$ Al $>$ Si

Beryllium (Be) is in period 2, while magnesium (Mg), aluminum (Al), and silicon (Si) are in period 3.

Since atomic radii increase with the number of electron shells, Be (with only 2 shells) should be **smaller** than Mg.

This means that stating Be $>$ Mg is incorrect.

Thus, the incorrect decreasing order of atomic radii is given in Option D.

Question29

Given below are two statements :

Statement I : According to the Law of Octaves, the elements were arranged in the increasing order of their atomic number.

Statement II : Meyer observed a periodically repeated pattern upon plotting physical properties of certain elements against their respective atomic numbers.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both **Statement I** and **Statement II** are false

B.

Statement I is false but **Statement II** is true

C.

Statement I is true but **Statement II** is false

D.

Both **Statement I** and **Statement II** are true

Answer: A

Solution:

Law of octaves was arranged in the increasing order of their atomic weight.

Lothar Meyer plotted the physical properties such as atomic volume, melting point and boiling point against atomic weight.

Question30

Given below are two statements :

Statement (I) : The radii of isoelectronic species increases in the order.

$Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

Statement (II) : The magnitude of electron gain enthalpy of halogen decreases in the order.

$Cl > F > Br > I$

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A.

Statement I is incorrect but Statement II is correct

B.

Both Statement I and Statement II are correct

C.

Statement I is correct but Statement II is incorrect

D.

Both Statement I and Statement II are incorrect

Answer: B

Solution:

Statement I : Correct

The radii of isoelectronic species increases in the order $\mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{F}^{-} < \mathrm{O}^{2-}$ (number of electrons = 10)

In general, radius of cation is smaller than anion $\mathrm{A}^{+} < \mathrm{A}^{-}$ cation with positive charge have higher effective nuclear charge.

Reason : In cations, number of electrons decreases, but number of protons remains the same. This causes a stronger pull on the remaining electrons by the nucleus, resulting in a higher effective nuclear charge.

With increase in effective nuclear charge, radii decreases so, increase in positive charge indicate more number of electron removal and hence increase in effective nuclear charge. As a result, radii decreases. So, Mg^{2+} has lower radii than Na^{+}.

For anion, number of electrons are increased. The electron-electron repulsion weakens the attraction between the nucleus and the electrons, resulting in a decrease in effective nuclear charge. So, effective nuclear charge of O^{2-}$islowerthanF^{-}$.

So, O^{2-} with lower effective nuclear charge causes longer radii of it.

So, $F^{-} < O^{2-}$

So, $Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

Ionic radii of isoelectronic species decreases as nuclear charge increases. The cation with greater +ve charge will have a smaller radius and the anion with the greater $-$ve charge will have a bigger radius for isoelectronic species.

Statement II : Correct

The magnitude of electron gain enthalpy of halogen decreases in the order Cl > F > Br > I.

Electron gain enthalpy is the amount of energy released when an atom accepts the electron from any neutral isolated gaseous atom to form an anion.

Generally, electron gain enthalpy decreases down the group. The exception is for F and cl. Electron gain enthalpy of F is less than Cl. It is due to the comparatively small size of F atom.

For Cl, Br and I, the electron gain enthalpy decreases as go from Cl to I.

So, both the statements are correct.

Question31

An element 'E' has the ionisation enthalpy value of 374 kJ mol^{-1}. 'E' reacts with elements A, B, C and D with electron gain enthalpy values of −328, −349, −325 and −295 kJ mol^{-1}, respectively. The correct order of the products EA, EB, EC and ED in terms of ionic character is :

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Options:

A.

EB > EA > EC > ED

B.

EA > EB > EC > ED

C.

ED > EC > EA > EB

D.

ED > EC > EB > EA

Answer: A

Solution:

Element $E \rightarrow$ Ionisation enthalpy value $\rightarrow 374 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Element $A \rightarrow$ Electron gain enthalpy value $\rightarrow -328 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Element $B \rightarrow$ Electron gain enthalpy value $\rightarrow -349 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Element $C \rightarrow$ Electron gain enthalpy value $\rightarrow -325 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Element $D \rightarrow$ Electron gain enthalpy value $\rightarrow -295 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Ionization enthalpy is the energy required to remove an election from an atom or ion in the gas phase. An element with lower ionization enthalpy indicate easier cation formation and this lead to higher ionic character. Electron gain enthalpy is the energy released when an election is added to an atom or ion in the gas phase. A highly negative election gain enthalpy Indicate that the atom readily gains electrons, favouring the formation of anions. This leads to stronger ionic character in compounds. High ionic character generally occurs when the difference in ionisation enthalpy and electron gain enthalpy between the two atoms is large.

For EA,

Difference between ionisation enthalpy and electron gain enthalpy

$$\begin{aligned} & =374 \mathrm{kJmol}^{-1}-\left(-328 \mathrm{~kJ} \mathrm{~mol}^{-1}\right) \\ & =702 \mathrm{~kJ} \mathrm{~mol}^{-1}\end{aligned}$$

For EB,

Difference between ionisation enthalpy and elation gain enthalpy

$$\begin{aligned} & =374 \mathrm{~kJ} \mathrm{~mol}^{-1}-\left(-349 \mathrm{~kJ} \mathrm{~mol}^1\right) \\ & =723 \mathrm{~kJ} \mathrm{~mol}^{-1}\end{aligned}$$

For EC,

Difference between ionisation enthalpy and election gain enthalpy

$$\begin{aligned} & =374 \mathrm{~kJ} \mathrm{~mol}^{-1}-\left(-325 \mathrm{~kJ} \mathrm{~mol}^{-1}\right) \\ & =699 \mathrm{~kJ} \mathrm{~mol}^{-1}\end{aligned}$$

For ED,

Difference between ionisation enthalpy and election gain enthalpy

$$\begin{aligned} & =374 \mathrm{~kJ} \mathrm{~mol}^{-1}-\left(-295 \mathrm{~kJ} \mathrm{~mol}^{-1}\right) \\ & =669 \mathrm{~kJ} \mathrm{~mol}^{-1}\end{aligned}$$

Higher difference between ionisation enthalpy and election gain enthalpy is for compound EB. So, EB has higher ionic character.

The Lower value of difference between ionisation enthalpy and election gain enthalpy is for compound ED. So, ED has lower ionic character. The increasing order of compounds for the difference between ionisation enthalpy and electron gain enthalpy $\rightarrow$

$EB < EC < EA < EB$

So, increasing. order of compounds for ionic character:

$ED < EC < EA < EB$

In the decreasing order,

$EB > EA > EC > ED$

Correct option (1)

Question32

First ionisation enthalpy values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of apatite family :

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Options:

A.

834 kJ mol^{-1}

B.

947 kJ mol^{-1}

C.

1402 kJ mol^{-1}

D.

1012 kJ mol^{-1}

Answer: D

Solution:

Apalite family - A group of phosphate minerals that includes fluorapatite, chlorapatite, and hychoxyapatite. Chemical formula $Ca_{10}(PO_4)_6(OH, F, Cl)_2$

$$\left.\begin{array}{l} Fluorapatite \rightarrow Ca_{10}(PO_4)_6F_2 \\ Chlorapatite \rightarrow Ca_{10}(PO_4)_6Cl_2 \\ Hydroxyapatite \rightarrow Ca_{10}(PO_4)_6(OH)_2 \end{array}\right\} \begin{array}{c} Main\,components: \\ Calcium\,and\,phosphate \end{array}$$

So, the element considered is phosphorus.

The first four group 15 elements are Nitrogen (N), Phosphorus (P), Arsenic (As) and Antimony (Sb). The outer configuration of phosphorus is half filled and hence stable (s^2p^3).

For a stable configuration, ionization enthalpy is high. Ionization enthalpy - Energy required to remove an electron from the gaseous atom.

Nitrogen has the highest first ionization enthalpy in group 15.

Down the group, ionization enthalpy decreases.

Down the group, atomic size increases, valence electrons are less attracted to the nucleus. And here, electron removal will become easier. So, ionization enthalpy is also decreases.

The highest value of ionization enthalpy 1402 KJ mol^{-1} corresponds to nitrogen element.

So, the first ionization enthalpy of the element that is a main component of apatite family is option D) 1012 kJ mol^{-1}.

Question33

The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is :

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Options:

A.

AO

B.

AO_2

C.

A_2O

D.

A_2O_3

Answer: D

Solution:

The elements given are

Li, Na, Be, Mg, B and Al

2^{nd} period elements $\rightarrow$ Li, Be, B

Order of atomic radii $\rightarrow$ Li > Be > B

3^{rd} period elements $\rightarrow$ Na, Mg, Al

Order of atomic radii $\rightarrow$ Na > Mg > Al

In a period, atomic radii decreases on moving from left to right.

So, the element with least atomic radii in 2_{nd} period (Li, Be, B) is B. The element with least atomic radii 3_{rd} period (Na, Mg, Al) is Al.

In B and Al $\Rightarrow$ B and Al are same group elements. Down the group, radii increases. So, B has least radii.

B is the element with least atomic radius to form the oxide A_2O_3.

Oxidation states of the A in oxides :

$A_0 \rightarrow +2 \; A_{0_2} \rightarrow +2 \; A_{2^0} \rightarrow +1 \; A_{2^0_3} \rightarrow +3$

B outer configuration is s^2P^1. 3 valence electrons and form A_2O_3 type oxide.

Question34

The property/properties that show irregularity in first four elements of group-17 is/are

(A) Covalent radius

(B) Electron affinity

(C) Ionic radius

(D) First ionization energy

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. A, B, C and D

B. A and C only

C. B only

D. B and D only

Answer: C

Solution:

The order of first four elements of group-17 are as follows.

$F < Cl < Br < I$ (Covalent radius)

$Cl > F > Br > I$ (Electron affinity)

$F < Cl^- < Br^- < I$ (Ionic radius)

$F > Cl > Br > I$ (I^{st} ionization energy)

Electron affinity order is irregular.

Question35

Given below are two statements :

Statement (I) : The metallic radius of Al is less than that of Ga .

Statement (II) : The ionic radius of Al^{3+} is less than that of Ga^{3+}.

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. Statement I is correct but Statement II is incorrect

B. Both Statement I and Statement II are correct

C. Statement I is incorrect but Statement II is correct

D. Both Statement I and Statement II are incorrect

Answer: C

Solution:

$\Rightarrow$ The metallic radius order of Al&Ga is

$$\mathrm{B} < \underbrace{\mathrm{Ga} < \mathrm{Al}}_{\downarrow} < \mathrm{In} < \mathrm{T}\ell$$

(due to poor shielding of d-subshell electrons)

$\Rightarrow$ The ionic radius order of Al^{+3}&Ga^{+3} is

$\mathrm{B}^{+3} < \mathrm{Al}^{+3} < \mathrm{Ga}^{+3} < \mathrm{In}^{+3} < \mathrm{T}\ell^{+3}$

Question36

Electronic configuration of four elements A, B, C and D are given below :

(A) $1s^2 2s^2 2p^3$

(B) $1s^2 2s^2 2p^4$

(C) $1s^2 2s^2 2p^5$

(D) $1s^2 2s^2 2p^2$

Which of the following is the correct order of increasing electronegativity (Pauling's scale)?

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. $\mathrm{D}<\mathrm{A}<\mathrm{B}<\mathrm{C}$

B. $\mathrm{A}<\mathrm{B}<\mathrm{C}<\mathrm{D}$

C. $\mathrm{A}<\mathrm{C}<\mathrm{B}<\mathrm{D}$

D. $\mathrm{A}<\mathrm{D}<\mathrm{B}<\mathrm{C}$

Answer: A

Solution:

The electronic configurations of four elements, A, B, C, and D, are presented as follows:

(A) $1s^2 2s^2 2p^3$

(B) $1s^2 2s^2 2p^4$

(C) $1s^2 2s^2 2p^5$

(D) $1s^2 2s^2 2p^2$

To determine the correct order of increasing electronegativity according to Pauling's scale, analyze the elements as follows:

Nitrogen ($1s^2 2s^2 2p^3$): Electronegativity = 3.0

Oxygen ($1s^2 2s^2 2p^4$): Electronegativity = 3.5

Fluorine ($1s^2 2s^2 2p^5$): Electronegativity = 4.0

Carbon ($1s^2 2s^2 2p^2$): Electronegativity = 2.55

The correct order of increasing electronegativity is:

Carbon (D) < Nitrogen (A) < Oxygen (B) < Fluorine (C)

Question37

Which of the following statements are correct?

A. The process of adding an electron to a neutral gaseous atom is always exothermic.

B. The process of removing an electron from an isolated gaseous atom is always endothermic.

C. The 1^{st} ionization energy of boron is less than that of beryllium.

D. The electronegativity of C is 2.5 in CH_4 and CCl_4

E. Li is the most electropositive among elements of group I.

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. A, C and D Only

B. B and C Only

C. B and D Only

D. B, C and E Only

Answer: B

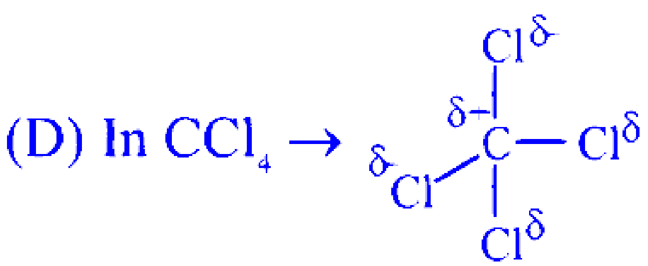
Solution:

(A) The process of adding an e^- to a neutral gaseous atom is not always exothermic it may be exothermic or endothermic.

(C) Be B
$1s^2 2s^2$ $1s^2 2s^2 2p^1$

In Be 2s subshell in fully filled

So, need high energy to remove e^- as compared to B .



due to partially positive charge z_{eff} $\uparrow$, EN $\uparrow$

So, EN of C $\Rightarrow CCl_4 > CH_4$

(E) Cs is most electropositive.

Question38

Match the LIST-I with LIST-II

LIST-I (Family)		LIST-II (Symbol of Element)	
A.	Pnictogen (group 15)	I	Ts
B	Chalcogen	II	Og
C	Halogen	III	Lv
D	Noble gas	IV	Mc

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. A-IV, B-I, C-II, D-III

B. A-IV, B-III, C-I, D-II

C. A-III, B-I, C-IV, D-II

D. A-II, B-III, C-IV, D-I

Answer: B

Solution:

To correctly match the elements with their respective families and symbols, consider the following associations:

Pnictogen is matched with Moscovium (Mc), which has the atomic number 115.

Chalcogen corresponds to Livermorium (Lv), with the atomic number 116.

Halogen is associated with Tennessine (Ts), with atomic number 117.

Noble gas correlates with Oganesson (Og), which has an atomic number of 118.

This alignment clarifies the classification of elements based on their family groups and chemical symbols.

Question39

The correct orders among the following are

Atomic radius : $\mathrm{B} < \mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl}$

Electronegativity : $\mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl} < \mathrm{B}$

Density : $\mathrm{Tl} < \mathrm{In} < \mathrm{Ga} < \mathrm{Al} < \mathrm{B}$

1st Ionisation Energy : $\mathrm{In} < \mathrm{Al} < \mathrm{Ga} < \mathrm{Tl} < \mathrm{B}$

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. B and D Only

B. A and B Only

C. C and D Only

D. A and C Only

Answer: A

Solution:

-	B	Al	Ga	In	Tl
Atomic radius (pm)	88	143	135	167	170
Electronegativity	2	1.5	1.6	1.7	1.8
Density (g/cm3)	2.35	2.7	5.9	7.31	11.85
Ionisation Energy (kJ/mol)	801	577	579	558	589

Radius Order $\mathrm{T\ell > In > A\ell > Ga > B}$

EN Order $\mathrm{B > T\ell > In > Ga > Al}$

Density Order $\mathrm{T\ell > In > Ga > A\ell > B}$

$\mathrm{IE_1}$ Order $\mathrm{B > T\ell > Ga > A\ell > In}$

Question40

The elements of Group 13 with highest and lowest first ionisation enthalpies are respectively :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. B & Tl

B. Tl & B

C. B & In

D. B & Ga

Answer: C

Solution:

IE order
$B > T\ell > Ga > Al > In$

Question41

The incorrect relationship in the following pairs in relation to ionisation enthalpies is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. $Mn^{2+} < Fe^{2+}$

B. $Mn^{+} < Mn^{2+}$

C. $Mn^{+} < Cr^{+}$

D. $Fe^{2+} < Fe^{3+}$

Answer: A

Solution:

Mn^{2+} : $[Ar]3\ d^5$

Half filled stability

More 1E than Fe^{2+}

Fe^{2+} : $[Ar]3\ d^6$

Question42

The group 14 elements A and B have the first ionisation enthalpy values of 708 and $715\ kJ\ mol^{-1}$ respectively. The above values are lowest among their group members. The nature of their ions A^{2+} and B^{4+} respectively is

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. both reducing

B. oxidising and reducing

C. both oxidising

D. reducing and oxidising

Answer: D

Solution:

Given the ionization energy values for elements A and B in group 14, where A has an ionization energy of 708 kJ/mol and B has 715 kJ/mol, these values are the lowest among their group. This helps identify the elements as tin (Sn) and lead (Pb), respectively. In this context:

The A^{2+} ion corresponds to Sn^{2+}, which acts as a reducing agent.

The B^{4+} ion corresponds to Pb^{4+}, which acts as an oxidizing agent.

Therefore, Sn in the +2 oxidation state tends to donate electrons (reduce), while Pb in the +4 oxidation state tends to accept electrons (oxidize), underscoring their respective roles as reducing and oxidizing agents.

Question43

The number of valence electrons present in the metal among $\mathrm{Cr}, \mathrm{Co}, \mathrm{Fe}$ and Ni which has the lowest enthalpy of atomisation is :

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. 10

B. 6

C. 9

D. 8

Answer: B

Solution:

To determine the number of valence electrons in the metal with the lowest enthalpy of atomization among Chromium (Cr), Cobalt (Co), Iron (Fe), and Nickel (Ni), we need to examine their respective electron configurations.

Chromium has the electron configuration:

$\mathrm{Cr}=[\mathrm{Ar}] 3 d^5 4 s^1$

Based on this configuration, Chromium has a total of six valence electrons (1 from the 4s orbital and 5 from the 3d orbital).

Chromium is noted for having the lowest enthalpy of atomization among the given metals. Thus, the number of valence electrons in Chromium, the metal with the lowest enthalpy of atomization, is 6.

Question44

Choose the incorrect trend in the atomic radii (r) of the elements.

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

$r_{\mathrm{Rb}} < r_{\mathrm{Cs}}$

B.

$r_{\mathrm{At}} < r_{\mathrm{Cs}}$

C.

$r_{\mathrm{Br}} < r_{\mathrm{K}}$

D.

$r_{\mathrm{Mg}} < r_{\mathrm{Al}}$

Answer: D

Solution:

In a period (row on the periodic table), the atomic size generally decreases as you move from left to right. This is because additional electrons are being added to the same outer shell while the number of protons in the nucleus increases, pulling the electron cloud closer to the nucleus and reducing the atomic radii.

Question45

Given below are two statements:

Statement I: H_2Se is more acidic than H_2Te

Statement II: H_2Se has higher bond enthalpy for dissociation than H_2Te

In the light of the above statements, choose the correct answer from the options given below:

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

Both Statement I and Statement II are true

B.

Statement I is true but Statement II is false

C.

Both Statement I and Statement II are false

D.

Statement I is false but Statement II is true

Answer: D

Solution:

To determine the relative acidity and bond enthalpy of H_2Se and H_2Te, let's examine both:

Acidic Strength:

The acidity of hydrides increases as we move down the group in the periodic table. This is because the bond strength between hydrogen and the central atom decreases, making it easier for the hydrogen ions to dissociate. Therefore, H_2Te, being further down the group than H_2Se, is more acidic:

$\mathrm{H_2Se} < \mathrm{H_2Te}$

Bond Enthalpy:

Bond enthalpy refers to the energy required to dissociate a bond. The bond enthalpy decreases down the group due to weaker bonds forming as the atomic size increases. Hence, H_2Se has a higher bond enthalpy compared to H_2Te:

$\Delta_{\text{dis}}\ \mathrm{H} : \mathrm{H_2Se} > \mathrm{H_2Te}$

$\mathrm{H_2Se}\,[276\ \mathrm{KJ/mol}] > \mathrm{H_2Te}\,[238\ \mathrm{KJ/mol}]$

Thus, H_2Se is indeed less acidic than H_2Te, but it has a higher bond enthalpy for dissociation.

Question46

The atomic number of the element from the following with lowest 1^{st} ionisation enthalpy is :

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

32

B.

35

C.

19

D.

87

Answer: D

Solution:

To determine which element has the lowest first ionization enthalpy among the given options, we look at the atomic numbers:

Atomic number 32 corresponds to Germanium (Ge).

Atomic number 35 corresponds to Bromine (Br).

Atomic number 87 corresponds to Francium (Fr).

Atomic number 19 corresponds to Potassium (K).

The element with the lowest first ionization energy is Francium (Fr) with atomic number 87. This can be represented by its electron configuration: $\mathrm{Fr} - [\mathrm{Rn}]7s^1$. Francium, being a part of the alkali metal group, has the most loosely bound outer electron compared to the other elements listed, resulting in a lower first ionization enthalpy.

Question47

Inert gases have positive electron gain enthalpy. Its correct order is [25-Jan-2023 Shift 1]

Options:

A. Xe < Kr < Ne < He

B. He < Ne < Kr < Xe

C. He < Xe < Kr < Ne

D. He < Kr < Xe < Ne

Answer: C

Solution:

Element	$\Delta egH[KJ/mol]$
He	+48
Ne	+116
Kr	+96
Xe	+77

From NCERT
So, order is $Ne > Kr > Xe > He$

Question48

Which of the following represents the correct order of metallic character of the given elements?
[25-Jan-2023 Shift 2]

Options:

A. $Si < Be < Mg < K$

B. $Be < Si < Mg < K$

C. $K < Mg < Be < Si$

D. $Be < Si < K < Mg$

Answer: A

Solution:

Metallic character increases down the group and decreases along the period.

Question49

The bond dissociation energy is highest for
[29-Jan-2023 Shift 1]

Options:

A. Cl_2

B. I_2

C. Br_2

D. F_2

Answer: A

Solution:

Bond energy of F_2 less than Cl_2 due to lone pair lone pair repulsions.
Bond energy order $Cl_2 > Br_2 > F_2 > I_2$

Question50

Match List - I with List - II

LIST-I (Atomic number)	LIST-II (Block of periodictable)
(A) 37	I. p-block
(B) 78	II. d-block
(C) 52	III. f-block
(D) 65	IV. s-block

Choose the correct answer from the options given below: [30-Jan-2023 Shift 1]

Options:

A. A - II, B - IV, C - I, D - III

B. A - I, B - III, C - IV, D - II

C. A - IV, B - III, C - II, D - I

D. A - IV, B - II, C - I, D - III

Answer: D

Solution:

Atomic number	Block
37(K)	s-block
78(Pt)	d-block
52(Te)	p-block
65(Tb)	f-block

Question51

The correct increasing order of the ionic radii is [31-Jan-2023 Shift 1]

Options:

A. $Cl^{-} < Ca^{2+} < K^{+} < S^{2-}$

B. $K^{+} < S^{2-} < Ca^{2+} < Cl^{-}$

C. $S^{2-} < Cl^{-} < Ca^{2+} < K^{+}$

D. $Ca^{2+} < K^{+} < Cl^{-} < S^{2-}$

Answer: D

Solution:

In isoelectronic species size $\propto \frac{1}{Z}$

$Ca^{2+} < K^{+} < Cl^{-} < S^{2-}$: Size

Z: 20 19 17 18

Z : 20191718

Question52

Nd^{2+} = ________ [31-Jan-2023 Shift 1]

Options:

A. $4f^{2}6s^{2}$

B. $4f^{4}$

C. $4f^3$

D. $4f^4 6s^2$

Answer: B

Solution:

$Nd(60) = [Xe]4f^4 5d^0 6s^2$
$Nd^{2+} = [Xe]4f^4 5d^0 5s^0$

Question53

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A) : The first ionization enthalpy of 3d series elements is more than that of group 2 metals
Reason (R): In 3d series of elements successive filling of d-orbitals takes place.
In the light of the above statements, choose the correct answer from the options given below :
[31-Jan-2023 Shift 2]

Options:

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true but (R) is not the correct explanation of (A)

C. (A) is false but (R) is true

D. (A) is true but (R) is false

Answer: C

Solution:

From Sc to Mn ionization energy is less than that of Mg
For 3 d series:

	Sc	Ti	V	Cr	Mn
IE(KJ / mol)	631	656	650	653	717
	Fe	Co	Ni	Cu	Zn
IE(KJ / mol)	762	758	736	745	906

For 2 ^{ad} Group

	Be	Mg	Ca	Sr	Ba	Ra
IE(KJ / mol)	631	656	650	653	717	762

Question54

Which of the following elements have half-filled f-orbitals in their ground state?
(Given : atomic number
Sm = 62; Eu = 63; Tb = 65; Gd = 64, Pm = 61)
A. Sm
B. Eu
C. Tb
D. Gd
E. Pm
Choose the correct answer from the options given below:
[31-Jan-2023 Shift 2]

Options:

A. B and D only

B. A and E only

C. A and B only

D. C and D only

Answer: A

Solution:

1. $62\, Sm : 4f^6 6\, s^2$
2. ${}_{64}Gd : 4f^7 5d^1 6\, s^2$
3. ${}_{63}Eu : 4f^7 6\, s^2$
4. ${}_{65}Tb : 4f^9 6\, s^2$
5. ${}_{61}Pm : 4f^5 6\, s^2$

Question55

For electron gain enthalpies of the elements denoted as $\Delta_{eg}H$, the incorrect option is :
[1-Feb-2023 Shift 2]

Options:

A. $\Delta_{eg}H(Cl) < \Delta_{eg}H\ (F)$

B. $\Delta_{eg}H(Se) < \Delta_{eg}H(S)$

C. $\Delta_{eg}H\ \ (I) < \Delta_{eg}H\ \ (At)$

D. $\Delta_{eg}H(Te) < \Delta_{eg}H(Po)$

Answer: B

Solution:

(1) $\Delta_{eg}H(Cl) < \Delta_{eg}H(F)$
$(-345)(-328)$ Correct
(2) $\Delta_{eg}H(Se) < \Delta_{eg}H(S)$
$(-195)\ \ (-200)$ Incorrect
(3) $\Delta_{eg}H$ (I) $<\Delta_{eg}H$ (At)
(-295) (-270) Correct
(4) $\Delta_{eg}H(Te) < \Delta_{eg}H(Po)$
$(-190)\ \ (-183)$ Correct

Question56

Which one of the following elements will remain as liquid inside pure boiling water ?

[6-Apr-2023 shift 2]

Options:

A. Cs

B. Ga

C. Li

D. Br

Answer: B

Solution:

Li, Cs reacts vigorously with water.
Br_2 changes in vapour state in boiling water $(BP = 58^{\circ}C)$
Ga reacts with water above $100^{\circ}C(MP = 29^{\circ}C, BP. = 2400^{\circ}C)$

Question57

Group-13 elements react with O_2 in amorphous form to form oxides of type M_2O_3 (M = element). Which among the following is the most basic oxide? [6-Apr-2023 shift 2]

Options:

A. Al_2O_3

B. Ga_2O_3

C. Tl_2O_3

D. B_2O_3

Answer: C

Solution:

As electropositive character increases basic character of oxide increases.
$\underset{\text{acidic}}{B_2O_3} < \underset{\text{amphoteric}}{Al_2O_3 < Ga_2O_3} < \underset{\text{basic}}{In_2O_3 < Tl_2O_3}$

Question58

The correct order of electronegativity for given elements is:
[8-Apr-2023 shift 1]

Options:

A. P > Br > C > At

B. C > P > At > Br

C. Br > P > At > C

D. Br > C > At > P

Answer: D

Solution:

C(2.5)
P(2.1) $\Rightarrow$ Br > C > At > P
Br(2.8)
At (2.2)

Question59

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A : The energy required to form Mg^{2+} from Mg is much higher than that required to produce Mg^{+}
Reason R: Mg^{2+} is small ion and carry more charge than Mg^{+}
In the light of the above statements, choose the correct answer from the options given below.
[10-Apr-2023 shift 2]

Options:

A. Both A and R are true and R is the correct explanation of A

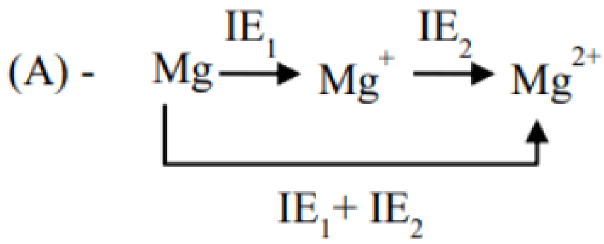
B. A is true but R is false

C. A is false but R is true

D. Both A and R are true but R is NOT the correct explanation of A

Answer: A

Solution:



In formation of Mg^{2+} $IE_1 + IE_2$ is required while in formation of Mg^{+} IE_1 is required
(R) Mg^{2+} is small ion and carry more change than $Mg^{\oplus}$

Question60

The correct order of metallic character is = [10-Apr-2023 shift 2]

Options:

A. $K > Be > Ca$

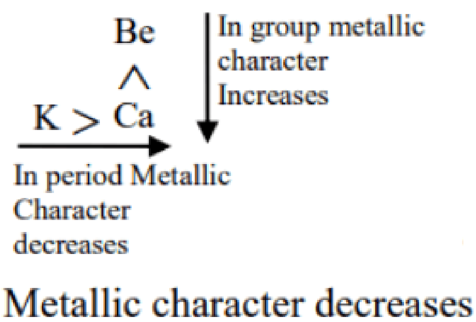
B. $Be > Ca > K$

C. $K > Ca > Be$

D. $Ca > K > Be$

Answer: C

Solution:



So $K > Ca > Be$

Question61

For compound having the formula $GaAlCl_4$, the correct option form the following is [11-Apr-2023 shift 1]

Options:

A. Cl forms bond with both Al and Ga in $GaAlCl_4$

B. Ga is coordinated with Cl in $GaAlCl_4$

C. Ga is more electronegative than Al and is present as a cationic part of the salt

D. Oxidation state of Ga in the salt $GaAlCl_4$ is +3

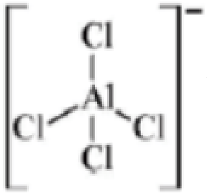
Answer: C

Solution:

Gallous tetrachloro aluminate $Ga^{+}AlCl_4^{-}$

$2\,Ga + Ga^{+}Cl_4^{-} + 2Al_2Cl_6 \xrightarrow{190^{\circ}} 4Ga^{+}AlCl_4^{-}$

Structure of $Ga^{+}AlCl_4^{-}$



Ga is cationic part of salt $GaAlCl_4$.

Question62

For elements B, C, N, Li, Be, O and F, the correct order of first ionization enthalpy is
[11-Apr-2023 shift 1]

Options:

A. B > Li > Be > C > N > O > F

B. Li < Be < B < C < N < O < F

C. Li < Be < B < C < O < N< F

D. Li< B < Be < C < O < N < F

Answer: D

Solution:

First I.E.

$F > N > O > C > Be > B > Li$

$Li - 520\,kJ/mol$

$Be - 899\,kJ/mol$

$B - 801\,kJ/mol$

$C - 1086\,kJ/mol$

$N - 1402\,kJ/mol$

$O - 1314 \, kJ / mol$
$F - 1681 \, kJ / mol$

Question63

Which of the following statements are not correct?
A. The electron gain enthalpy of F is more negative than that of Cl.
B. Ionization enthalpy decreases in a group of periodic table.
C. The electronegativity of an atom depends upon the atoms bonded to it.
D. Al_2O_3 and NO are examples of amphoteric oxides.
Choose the most appropriate answer from the options given below : [13-Apr-2023 shift 1]

Options:

A. A, C and D Only

B. B and D Only

C. A, B and D Only

D. A, B, C and D

Answer: A

Solution:

Electronegativity of an element depends on the atom with which it is attached.
$NO =$ neutral oxide
$Al_2O_3 =$ amphoteric oxide

Question64

Identify the correct order of standard enthalpy of formation of sodium halides. [13-Apr-2023 shift 2]

Options:

A. NaI < NaBr < NaF < NaCl

B. NaF < NaCl < NaBr < Nal

C. NaCl < NaF < NaBr < Nal

D. Nal < NaBr < NaCl < NaF

Answer: D

Solution:

For a given metal $\Delta_f H^0$ always becomes less negative from fluoride to iodide.

Question65

The number of molecules from the following which contain only two lone pair of electrons is ________
H_2O, N_2, CO, XeF_4, NH_3, NO, CO_2, F_2
[10-Apr-2023 shift 2]

Answer: 4

Solution:

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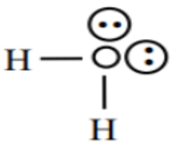
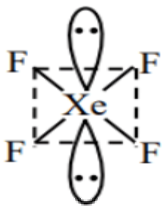
	2
$:N \equiv N:$	2
$:C \equiv O:$	2
	2
$\ddot{N}H_3$	1
$.N \equiv \ddot{\underset{..}{O}}$	3

Question66

Metals generally melt at very high temperature. Amongst the following, the

metal with the highest melting point will be
[24-Jun-2022-Shift-2]

Options:

A. Hg

B. Ag

C. Ga

D. Cs

Answer: B

Solution:

Melting points of the given metals
Hg : $-38.83^{\circ}C$
Ag : $961.8^{\circ}C$
Ga : $29.76^{\circ}C$
Cs : $28.44^{\circ}C$
∴ Metal having highest melting point is Ag.

Question67

The correct order of electron gain enthalpies of Cl, F, Te and Po is [25-Jun-2022-Shift-2]

Options:

A. F < Cl < Te < Po

B. Po < Te < F < Cl

C. Te < Po < Cl < F

D. Cl < F < Te < Po

Answer: B

Solution:

$Te \rightarrow -190 \text{ kJ mol}^{-1}$
$Po \rightarrow -174 \text{ kJ mol}^{-1}$
$F \rightarrow -333 \text{ kJ mol}^{-1}$
$Cl \rightarrow -349 \text{ kJ mol}^{-1}$
Hence, correct order is $Cl > F > Te > Po$

Question68

Which of the following elements is considered as a metalloid? [26-Jun-2022-Shift-2]

Options:

A. Sc

B. Pb

C. Bi

D. Te

Answer: D

Solution:

Sc, Pb, Bi are metals
Te is a metalloid

Question69

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The ionic radii of O^{2-} and Mg^{2+} are same.
Reason (R): Both O^{2-} and Mg^{2+} are isoelectronic species.
In the light of the above statements, choose the correct answer from the options given below.
[27-Jun-2022-Shift-1]

Options:

A. Both (A) and (R) are true and (R) is the correct explanation of (A).

B. Both (A) and (R) are true but (R) is not the correct explanation of (A).

C. (A) is true but (R) is false.

D. (A) is false but (R) is true.

Answer: D

Solution:

Correct order of ionic radii:
$O^{-2} > Mg^{+2}$
This is because among isoelectronic species, the size of anions are greater than the size of cations. Statement (II) is correct as both O^{-2} and Mg^{+2} are isoelectronic.

Question70

The correct order of increasing ionic radii is [27-Jun-2022-Shift-2]

Options:

A. $Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

B. $N^{3-} < O^{2-} < F^{-} < Na^{+} < Mg^{2+}$

C. $F^{-} < Na^{+} < O^{2-} < Mg^{2+} < N^{3-}$

D. $Na^{+} < F^{-} < Mg^{2+} < O^{2-} < N^{3-}$

Answer: A

Solution:

For isoelectronic species
Ionic radii $\propto \frac{(-)\ \text{ve charge}}{(+)\ \text{ve charge}}$
Hence, correct order of ionic radii is
$Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Question71

Element "E" belongs to the period 4 and group 16 of the periodic table. The valence shell electron configuration of the element, which is just above "E" in the group is
[28-Jun-2022-Shift-1]

Options:

A. $3s^2, 3p^4$

B. $3d^{10}, 4s^2, 4p^4$

C. $4d^{10}, 5s^2, 5p^4$

D. $2s^2, 2p^4$

Answer: A

Solution:

Element E is Selenium (Se)
Electronic configuration of E is $[Ar]3d^{10}4s^24p^4$
The element which is just above ' E ' in periodic table is sulphur, its electronic configuration is [Ne] $3s^23p^4$

Question72

Choose the correct answer from the options given below:

	List-I (Oxide)		List-II (Nature)
(A)	Cl_2O_7	(I)	Amphoteric
(B)	Na_2O	(II)	Basic
(C)	Al_2O_3	(III)	Neutral
(D)	N_2O	(IV)	Acidic

[28-Jun-2022-Shift-2]

Options:

A. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

B. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

C. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)

D. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)

Answer: B

Solution:

(A) $Cl_2O_7 \rightarrow$ Acidic
(B) $Na_2O \rightarrow$ Basic
(C) $Al_2O_3 \rightarrow$ Amphoteric
(D) $N_2O \rightarrow$ Neutral
Oxides of metals are basic in nature whereas oxides of non-metals are acidic in nature. N_2O is a neutral oxide.

Question73

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : The first ionization enthalpy for oxygen is lower than that of nitrogen.
Reason R : The four electrons in 2p orbitals of oxygen experience more electron-electron repulsion.
In the light of the above statements, choose the correct answer from the options given below.
[29-Jun-2022-Shift-2]

Options:

A. Both A and R are are correct and R is the correct explanation of A.

B. Both A and R are are correct but R is NOT the correct explanation of A.

C. A is correct but R is not correct.

D. A is not correct but R is correct.

Answer: B

Solution:

Nitrogen has half-filled p-orbitals which is stable. Due to this, its 1^{st} ionization energy is more than oxygen.

Question74

The IUPAC nomenclature of an element with electronic configuration $[Rn]5f^{14}6d^{1}7s^{2}$ is :
[25-Jul-2022-Shift-1]

Options:

A. Unnilbium

B. Unnilunium

C. Unnilquadium

D. Unniltrium

Answer: D

Solution:

The element with electronic configuration $[Rn]5f^{14}6d^{1}7s^{2}$ has atomic number $\rightarrow 103$
∴ Its IUPAC name is: Unniltrium

Question75

The first ionization enthalpies of Be, B, N and O follow the order [25-Jul-2022-Shift-2]

Options:

A. $0 < N < B < Be$

B. $Be < B < N < O$

C. $B < Be < N < O$

D. $B < Be < O < N$

Answer: D

Solution:

The first ionization energy increase from left to right along 2^{nd} period with the following exceptions
$IE_1 : Be > B$ and $N > O$
This is due to stable configuration of Be in comparison to B and that of N in comparison to O.
Hence the correct order is $N > O > Be > B$

Question76

Given two statements below :
Statement I : In Cl_2 molecule the covalent radius is double of the atomic radius of chlorine.
Statement II : Radius of anionic species is always greater than their parent atomic radius.

Choose the most appropriate answer from options given below :
[26-Jul-2022-Shift-1]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Answer: D

Solution:

Covalent radius is not double of atomic radius.
Radius of anionic species is always greater than their parent atomic radius as nuclear charge decreases in anionic counterpart.

Question77

Outermost electronic configurations of four elements A, B, C, D are given below :
(A) $3s^2$
(B) $3s^2 3p^1$
(C) $3s^2 3p^3$
(D) $3s^2 3p^4$
The correct order of first ionization enthalpy for them is :
[27-Jul-2022-Shift-2]

Options:

A. (A) < (B) < (C) < (D)

B. (B) < (A) < (D) < (C)

C. (B) < (D) < (A) < (C)

D. (B) < (A) < (C) < (D)

Answer: B

Solution:

Orbitals with fully filled and half-filled electronic configuration are stable, and require more energy for ionization
Elements with greater electronegativity require more energy for ionisation
Hence the correct order is C > D > A > B

Question78

In which of the following pairs, electron gain enthalpies of constituent elements are nearly the same or identical?
(A) Rb and Cs
(B) Na and K
(C) Ar and Kr
(D) I and At
Choose the correct answer from the options given below:
[28-Jul-2022-Shift-1]

Options:

A. (A) and (B) only

B. (B) and (C) only

C. (A) and (C) only

D. (C) and (D) only

Answer: C

Solution:

Rb& Cs have nearly same electron gain enthalpy electron gain enthalpy $= -46\,kj/ml$
Ar \& Kr have same ΔH_{eq}. Value is $+96\,kj/ml$

Question79

The correct decreasing order for metallic character is
[28-Jul-2022-Shift-2]

Options:

A. Na > Mg > Be > Si > P

B. P > Si > Be > Mg > Na

C. Si > P > Be > Na > Mg

D. Be > Na > Mg > Si > P

Answer: A

Solution:

Metallic character increases top to bottom in group and decreases left to right in a period.
Mg is from second group it will be less metallic than Na. Be comes above Mg hence less metallic than Mg. Si is more metallic than phosphorous.

Question80

Which of the following pair of molecules contain odd electron molecule and an expanded octet molecule?
[29-Jul-2022-Shift-1]

Options:

A. BCl_3 and SF_6

B. NO and H_2SO_4

C. SF_6 and H_2SO_4

D. BCl_3 and NO

Answer: B

Solution:

(A) BCl_3 - Even electron molecules
SF_6 - Expended octet molecules
(B) NO - Odd electron molecules H_2SO_4 - Expanded octet
(C) SF_6 - Even electron molecules H_2SO_4 - Expanded octet
(D) BCl_3 - Even electron molecules NO - odd electron molecules

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Question81

The first ionization enthalpy of Na, Mg and Si, respectively, are : 496, 737 and 786 $\mathrm{kJ\,mol^{-1}}$. The first ionization enthalpy ($\mathrm{kJ\,mol^{-1}}$) of Al is :
[29-Jul-2022-Shift-1]

Options:

A. 487

B. 768

C. 577

D. 856

Answer: C

Solution:

I.E. $Na < Al < Mg < Si$
$\because 496 <$ I.E. $(Al) < 737$
Option (C), matches the condition
i.e. I.E. $(Al) = 577\ kJ\ mol^{-1}$

Question82

Which pair of oxides is acidic in nature?
[26 Feb 2021 Shift 2]

Options:

A. B_2O_3, CaO

B. B_2O_3, SiO_2

C. N_2O, BaO

D. CaO, SiO_2

Answer: B

Solution:

(a) $B_2O_3 \longrightarrow$ oxide of boron (non-metal) :acidic
$CaO \longrightarrow$ oxide of calcium (metal) : basic
(b) $B_2O_3 \longrightarrow$ oxide of boron (non-metal) : acidic
$SiO_2 \longrightarrow$ oxide of silicon (non-metal) : acidic
(c) $N_2O \longrightarrow$ oxide of nitrogen (non-metal) : neutral
$BaO \longrightarrow$ oxide of barium (metal) : basic
(d) $CaO \longrightarrow$ oxide of calcium (metal): basic
$SiO_2 \longrightarrow$ oxide of silicon (non-metal): acidic
So, option (b) is the correct answer. B_2O_3 and SiO_2 are being acidic oxide they react with base to give salt and water.

$$B_2O_3 + 2NaOH \longrightarrow \underset{\text{Sodium metaborate}}{2NaBO_2} + H_2O$$

$$SiO_2 + 2NaOH \longrightarrow \underset{\text{Sodium silicate}}{Na_2SiO_3} + H_2O$$

Question83

Match List-I with List-II.

List-I(Electronicconfiguration ofelements)	List-II (Δ_i in $kJ\,mol^{-1}$)
A. $1s^2 2s^2$	(i) 801
B. $1s^2 2s^2 2p^4$	(ii) 899
C. $1s^2 2s^2 2p^3$	(iii) 1314
D. $1s^2 2s^2 2p^1$	(iv) 1402

Choose the most appropriate answer from the options given below. [26 Feb 2021 Shift 1]

Options:

A. (A-ii), (b-iii), (c-iv), (d-i)

B. (A-i), (b-iv), (c-iii), (d-ii)

C. (A-i), (b-iii), (c-iv), (d-ii)

D. (A-iv), (b-i), (c-ii), (d-iii)

Answer: A

Solution:

Here, (B), (C) and (D) are p-block elements of the 2 nd period.
(D) $\longrightarrow p^1$ configuration $\rightarrow$ B of group 13
(C) $\longrightarrow p^3$ configuration $\rightarrow$ N of group 15
(B) $\longrightarrow p^4$ configuration $\rightarrow$ O of group 16

Stability order : $\underbrace{p^3}_{\text{Half-filled}} > \underbrace{p^4 >> p^1}_{\text{Partially filled}}$

We know, ionisation enthalpy $(\Delta_i H) \propto$ stability of the subshell concerned.
Therefore, half-filled subshell is more stable than partially filled.
$\therefore \Delta_i H$ order is $c > b >> d$

(A) is a s-block element (group 2) of 2nd period with s^2-configuration $\rightarrow$ Be of group 2 [fully-filled; stable]
So, the correct order of IE, or $\Delta_i H_1$ (in $kJ\,mol^{-1}$) of the 2 nd period elements will be

$$\begin{array}{ccccccc} B & < & Be & \ll & O & < & N \\ (d) & & (a) & & (b) & & (c) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ (i) & & (ii) & & (iii) & & (iv) \\ [801] & & [899] & & [1319] & & [1902] \end{array}$$

The correct matching is (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)
Note The order of IE, or $\Delta_i H_1$ of 2 nd period elements is

$Li < B < Be < C < O < N < F < Ne$

Question84

Consider the elements Mg, Al, S, P and Si, the correct increasing order of their first ionization enthalpy is :
[24feb2021shift1]

Options:

A. $Mg < Al < Si < S < P$

B. $Al < Mg < Si < S < P$

C. $Mg < Al < Si < P < S$

D. $Al < Mg < S < Si < P$

Answer: B

Solution:

Mg and P exhibit abnormal behaviour. Due to extra stability of half-filled and fullfilled electronic configuration. First ionisation enthalpy order is $Al < Mg < Si < S < P$

Question85

AX is a covalent diatomic molecule, where A and X are second row elements of periodic table. Based on molecular orbital theory, the bond order of AX is 2.5. The total number of electrons in AX is .... (Round off to the nearest integer).
[18 Mar 2021 Shift 1]

Answer: 15

Solution:

AX is a covalent diatomic molecule. The molecule is NO (Nitric oxide). Its bond order is 2.5 and it has a total of 15 electrons $(7 + 8 = 15)$.
Here ' A ' denotes nitrogen (N) and ' X ' denotes oxygen (O). NO is colourless and toxic gas.
Note : Total number of electrons equal to 13 will also have the 2.5 bond order, but in this case neutral diatomic molecule will not be possible.

Question86

The set of elements that differ in mutual relationship from those of the other sets is
[17 Mar 2021 Shift 2]

Options:

A. Li - Mg

B. B-Si

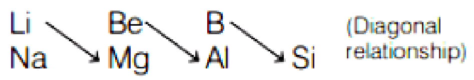
C. Be - Al

D. Li-Na

Answer: D

Solution:

Li-Na pair is different from remaining three options as it do not shows diagonal relationship. Both Li-Na belongs to same group and are not placed diagonally.
$Li-Mg$, $B-Si$ and $Be-Al$ show diagonal relationship.



Question87

Identify the elements X and Y using the ionisation energy values given below.

	Ionisation energy (Ist)	kJ/mol (IInd)
X	495	4563
Y	71	1450

[18 Mar 2021 Shift 1]

Options:

A. $X = Na, Y = Mg$

B. $X = Mg, Y = F$

C. $X = Mg, Y = Na$

D. $X = F, Y = Mg$

Answer: A

Solution:

Given data,

	IE_1	IE_2
X	495	4563 kJ/mol
Y	71	1450 kJ/mol

IE_1 IE_2
X 495 4563 kJ/mol
Y 71 1450 kJ/mol

X is $Na(3s^1)$ as it has very high second ionisation energy due to removal of electron from highly stable noble gas configuration.

$$\underset{1s^2 2s^2 2p^6 3s^1}{Na} \xrightarrow{IE_1} \underset{1s^2 2s^2 2p^6 \text{ or } [Ne]}{Na^+} + e^-; \; IE_1 = 495 kJ/mol$$

$$\underset{1s^2 2s^2 2p^6}{Na^+} \xrightarrow{IE_2} \underset{1s^2 2s^2 2p^5}{Na^{2+}} + e^-; IE_2 = 4563 kJ/mol$$

(Stable noble gas configuration).

In case of Mg, noble gas configuration is achieved after removal of second electron. So, IE_2 is comparatively less.

$$\underset{[Ne]3s^2}{Mg} \longrightarrow \underset{[Ne]3s^1}{Mg^+} + e^-; IE_1 = 731 kJ/mol$$

$$\underset{[Ne]3s^1}{Mg^+} \longrightarrow \underset{[Ne]}{Mg^{2+}} + e^-; IE_2 = 1450 kJ/mol$$

Question88

The first ionisation energy of magnesium is smaller as compared to that of elements X and Y, but higher than that of Z. The elements X, Y and Z, respectively, are
[18 Mar 2021 Shift 2]

Options:

A. chlorine, lithium and sodium

B. argon, lithium and sodium

C. argon, chlorine and sodium

D. neon, sodium and chlorine

Answer: C

Solution:

The order of first ionisation energy of 3nd period is as follows

$Na < Al < Mg < Si < S < P < Cl < Ar$

On moving along a period from left to right, nuclear charge increases that outweighs the shielding effect. As a result, outermost electrons are held more tightly. Hence, ionisation enthalpy increases.
Ionisation enthalpy of group 2 is greater than group 13 and that of group 15 is greater than group 16. This is due to fully filled and half-filled orbital.
$\therefore X$, Y and Z are argon, chlorine and sodium respectively.

Question89

Outermost electronic configuration of a group 13 element, E , is $4s^2$, $4p^1$. The electronic configuration of an element of p-block period-five placed diagonally to element, E is :
[20 Jul 2021 Shift 2]

Options:

A. $[Kr]3d^{10}4s^2 4p^2$

B. $[Ar]3d^{10}4s^2 4p^2$

C. $[Xe]5d^{10}6s^2 6p^2$

D. $[Kr]4d^{10}5s^2 5p^2$

Answer: D

Solution:

The element E is Ga and the diagonal element of 5^{th} period is $_{50}Sn$ having outer electronic configuration will be $[Kr]5s^2 4d^{10}5p^2$.

Question90

The CORRECT order of first ionisation enthalpy is :
[27 Jul 2021 Shift 2]

Options:

A. $Mg < S < Al < P$

B. $Mg < Al < S < P$

C. $Al < Mg < S < P$

D. $Mg < Al < P < S$

Answer: C

Solution:

Mg Al P S $\rightarrow$ IE. order $\Rightarrow$ Al< Mg < S < P

Valence $[N_e]$: $\overset{Mg}{3s^2}$ $\overset{Al}{3s^2 3p^1}$ $\overset{P}{3s^2 3p^3}$ $\overset{S}{3s^2 3p^4}$

$\uparrow$ $\uparrow$

Full Filled Stable — Half Filled Stable

Question91

The ionic radii of F^- and O^{2-} respectively are 1.33 Å and 1.4Å, while the covalent radius of N is 0.74ÅThe correct statement for the ionic radius of N^{3-} from the following is : [25 Jul 2021 Shift 2]

Options:

A. It is smaller than F^- and N

B. It is bigger than O^{2-} and F^-

C. It is bigger than F^- and N , but smaller than of O^{2-}

D. It is smaller than O^{2-} and F^-, but bigger than of N

Answer: B

Solution:

F^-, O^{2-} and N^{3-} all are isoelectronic species in which N^{3-} have least number of protons due to which it's size increases as least nuclear attraction is experienced by the outer shell electrons.
Size order $N^{3-} > O^{2-} > F^-$

Question92

The correct order of ionic radii for the ions, P^{3-}, S^{2-}, Ca^{2+}, K^+, Cl^- is [27 Aug 2021 Shift 2]

Options:

A. $P^{3-} > S^{2-} > Cl^- > K^+ > Ca^{2+}$

B. $Cl^- > S^{2-} > P^{3-} > Ca^{2+} > K^+$

C. $P^{3-} > S^{2-} > Cl^{-} > Ca^{2+} > K^{+}$

D. $K^{+} > Ca^{2+} > P^{3-} > S^{2-} > Cl^{-}$

Answer: A

Solution:

The electronic configuration of given ions are as follows:
$K_{19} = 2, 8, 8, 1$ $K^{+} = 2, 8, 8$
$Ca_{20} = 2, 8, 8, 2$ $Ca^{2+} = 2, 8, 8$
$P_{15} = 2, 8, 5$ $P^{3-} = 2, 8, 8$
$S_{16} = 2, 8, 6$ $S^{2-} = 2, 8, 8$
$Cl_{17} = 2, 8, 7$ $Cl^{-} = 2, 8, 8$
As the effective nuclear charge increases from P to Ca, the atomic size decreases in the same order.
Order of ionic radii
$\underset{15}{P^{3-}} > \underset{16}{S^{2-}} > \underset{17}{Cl^{-}} > \underset{19}{K^{+}} > \underset{20}{Ca^{2+}}$
Therefore, option (a) is correct.

Question93

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) Metallic character decreases and non-metallic character increases on moving from left to right in a period.
Reason (R) It is due to increase in ionisation enthalpy and decrease in electron gain enthalpy, when one moves from left to right in a period.
In the light of the above statements, choose the most appropriate answer from the options given below.
[31 Aug 2021 Shift 1]

Options:

A. (A) is false but (R) is true.

B. (A) is true but (R) is false

C. Both (A) and (R) are correct and (R) is the correct explanation of (A).

D. Both (A) and (R) are correct but (R) is not the correct explanation of (A).

Answer: B

Solution:

On moving from left to right in periodic table, metallic character decreases while non-metallic character increases. This is due to decrease in ionisation enthalpy and increase in electron gain enthalpy when we move from left to right in periodic table.

Hence, (A) is true while (R) is false.
Hence, correct option is (b).

Question94

Identify the element for which electronic configuration in +3 oxidation state is [Ar] $3d^5$.
[1 Sep 2021 Shift 2]

Options:

A. Ru

B. Mn

C. Co

D. Fe

Answer: D

Solution:

Iron (Fe) has electronic configuration $[Ar]3d^5$ in +3 oxidation state.
Electronic configuration of $Fe = [Ar]3d^6, 4s^2$
Electronic configuration of $Fe^{3+} = [Ar]3d^5$.

Question95

B has a smaller first ionization cnthalpy than Be. Consider the following statements:
(I) it is casier to remove 2p electron than 2s electron
(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be
(III) 2s electron has more penetration power than 2p electron (IV) atomic radius of B is more than Be (atomic number B = 5, Bc = 4)
The correct statements are:
[Jan. 09,2020 (I)]

Options:

A. (I), (II) and (IV)

B. (II), (III) and (IV)

C. (I), (II) and (III)

D. (I), (III) and (IV)

Answer: C

Solution:

$_5B : 1s^2 2s^2 2p^1$
$_4Be : 1s^2 2s^2$
First ionisation enthalpy of B is lower than Be because Be has a stable electronic configuration. It required more energy to remove the first electron from $2s$(in Be) than $2p$ (in B) because $2se^-$ has more penetration power than $2p$. Therefore options (I), (II) and (III) are correct. Atomic radius of B is less than Be.

Question96

The acidic, basic and amphoteric oxides, respectively, are:
[Jan. 09,2020 (I)]

Options:

A. Na_2O, SO_3, A_2O_3

B. Cl_2O, CaO, P_4O_{10}

C. N_2O_3, Li_2O, Al_2O_3

D. MgO, Cl_2O, Al_2O_3

Answer: C

Solution:

Generally, non-metal oxides are acidic in nature and metal oxides are basic in nature, Al_2O_3 is amphoteric.

Question97

The first and second ionisation enthalpies of a metal are 496 and 4560kJ mol^{-1}, respectively. How many moles of HCl and H_2SO_4, respectively, will be needed to react completely with 1 mole of the metal hydroxide?
[NV, Jan. 09,2020(II)]

Options:

A. 1 and 1

B. 2 and 0.5

C. 1 and 2

D. 1 and 0.5

Answer: D

Solution:

A large difference between first and second ionisation enthalpies $(4560-496-4064 \mathrm{kJ\,mol}^{-1})$ confirms the metal to be an alkali metal and thus is monovalent and form hydroxide of the type $M(OH)$

$$\underset{1\,\mathrm{mol}}{MOH} + \underset{1\,\mathrm{mol}}{HCl} \rightarrow MCl + H_2O$$

$$\underset{1\,\mathrm{mol}}{2MOH} + H_2SO_4 \rightarrow M_2SO_4 + \underset{1/2\,\mathrm{mol}}{H_2O}$$

Question98

The first ionization energy (in kJ/mol) of Na, Mg, Al and Si respectively, are: [Jan. 08,2020(I)]

Options:

A. 496, 737, 577, 786

B. 496, 577, 737, 786

C. 786, 737, 577, 496

D. 496, 577, 786, 737

Answer: A

Solution:

All given elements belongs to period III and generally their ionisation energy will increase along the period but Mg will show higher ionisation potential compared to Al due to its stable configuration. Thus correct order of ionisation energy will be: $Na < Al < Mg < Si$.
Ionisation energy (kJ/mol) of the given metals are
$Na: 496; Al: 577; Mg: 737; Si: 786$

Question99

The increasing order of the atomic radii of the following elements is:
(i) C
(ii) O
(iii) F
(iv) Cl

(v) Br
[Jan. 08,2020 (II)]

Options:

A. (ii) <(iii) < (iv) < (i) < (v)

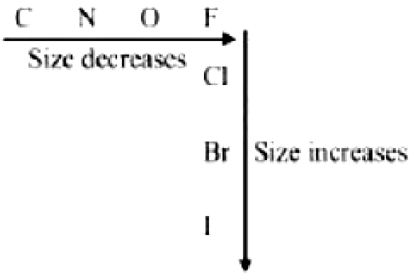
B. (iv) <(iii) < (ii) < (i) < (v)

C. (iii) <(ii) < (i) < (iv) < (v)

D. (i) < (ii) < (iii) < (iv) < (v)

Answer: C

Solution:



Correct increasing order of atomic radii is
$F < O < C < Cl < Br$

Question100

The electron gain enthalpy (in kJ / mol) of fluorine, chlorine, bromine and iodine, respectively, are:
[Jan. 07,2020 (I)]

Options:

A. −296, −325, −333 and −349

B. −349, −333, −325 and −296

C. −333, −349, −325 and −296

D. −333, −325, −349 and −296

Answer: C

Solution:

Chlorine has highest electron gain enthalpy (most negative) among the given elements, the electron gain enthalpy decreases down the group i.e., moves to least negative.

Question101

Within each pair of elements F & Cl, S & Se, and Li & Na, respectively, the elements that release more cnergy upon an electron gain are:
[Jan. 07, 2020 (II)]

Options:

A. Cl, Se and Na

B. Cl, S and Li

C. F, S and Li

D. F, Se and Na

Answer: B

Solution:

Down the group electron affinity decrease due to increasing size. So, the electron affinity of $S >$ Se and $Li >$ Na so energy release will also be more. But in case of F and Cl, there is exception, $Cl > F$. It is due to the small size of F atom where the incoming electron face high electron-electron repulsion. So, the energy release will be more in Cl.

Question102

The atomic number of Unnilunium is ________.
[NV,Sep. 06,2020(II)]

Answer: 101

Solution:

IUPAC symbol = Unu
Atomic no. $(Z) = 101$

Question103

The atomic number of the element unnilennium is:
[Sep. 03,2020 (I)]

Options:

A. 109

B. 102

C. 108

D. 119

Answer: A

Solution:

un $= 1$
nil $= 0$
enn $= 9$
So, atomic number $= 109$

Question104

The set that contains atomic numbers of only transition elements, is : [Sep. 06,2020(I)]

Options:

A. 37, 42, 50, 64

B. 21, 25, 42, 72

C. 9, 17, 34, 38

D. 21, 32, 53, 64

Answer: B

Solution:

Elements with atomic number 21, 25, 42 and 72 belongs to transition metals.

Question105

The correct order of the ionic radii of O^{2-}, N^{3-}, F^{-}, Mg^{2+}, Na^{+} and Al^{3+} is : [Sep. 05, 2020 (II)]

Options:

A. $N^{3-} < O^{2-} < F^{-} < Na^{+} < Mg^{2+} < Al^{3+}$

B. $Al^{3+} < Na^{+} < Mg^{2+} < O^{2-} < F^{-} < N^{3-}$

C. $Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

D. $N^{3-} < F^{-} < O^{2-} < Mg^{2+} < Na^{+} < Al^{3+}$

Answer: C

Solution:

All are isoelectronic species, so more is the Z_{eff} less will be the ionic size.
∴ Correct order of ionic radii is
$Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Question106

The elements with atomic numbers 101 and 104 belong to, respectively: [Sep .04,2020(I)]

Options:

A. Group 11 and Group 4

B. Actinoids and Group 6

C. Actinoids and Group 4

D. Group 6 and Actinoids

Answer: C

Solution:

${}_{89}AC \longrightarrow {}_{103}Lr$
Belongs to actinoids series and they all belongs to 3^{id} group. So atomic no. 101 element is actinoids and atomic number 104 element belongs to 4^{th} group.

Question107

The process that is NOT endothermic in nature is: [Sep. 04, 2020 (II)]

Options:

A. $Ar(g) + e^{-} \rightarrow Ar^{-}(g)$

B. $H(g) + e^{-} \rightarrow H^{-}(g)$

C. $O^{-}(g) + e^{-} \rightarrow O^{2-}(g)$

D. $Na(g) \longrightarrow Na^{+}(g) + e^{-}$

Answer: B

Solution:

$Ar(g) + e^{-} \longrightarrow Ar^{-}$ (g) (endothermic)
$H(g) + e^{-} \longrightarrow H^{-}$ (g) (exothermic)
$Na(g) \longrightarrow Na^{+}(g) + e^{-}$ (endothermic)
$O^{-}(g) + e^{-} \longrightarrow O^{2-}(g)$ (endothermic)
- Electron gaining enthalpy (EGE) of $H(g)$ is negative while that of $Ar(g)$ is positive due to ns^2np^6 configuration.
- Second EGE is always positive for an atom.
- Ionization potential of an atom is positive.

Question108

The ionic radii of O^{2-}, F^{-}, Na^{+} and Mg^{2+} are in the order: [Sep. 04,2020(I)]

Options:

A. $F^{-} > O^{2->} > Na^{+} > Mg^{2+}$

B. $O^{2->} F^{->} Na^{+} > Mg^{2+}$

C. $Mg^{2+} > Na^{+} > F^{-} > O^{2}$

D. $O^{2-} > F^{-} > Mg^{2+} > Na^{+}$

Answer: B

Solution:

	O^{2-}	F^{-}	Na^{+}	Mg^{2+}
Z	8	9	11	12
No. of e^{-}	10	10	10	10

In isoelectronic species greater is $Z_{effe.}$ smaller is radius so order is $O^{2-} > F^{-} > Na^{+} > Mg^{2+}$.

Question109

Among the statements (I - IV), the correct ones are:
(I) Be has smaller atomic radius compared to Mg.

(II) Be has higher ionization enthalpy than Al .
(III) Charge\/radius ratio of Be is greater than that of Al.
(IV) Both Be and Al form mainly covalent compounds.
[Sep. 03,2020 (II)]

Options:

A. (I), (II) and (IV)

B. (I), (III) and (IV)

C. (II),(III) and(IV)

D. (I), (II) and (III)

Answer: A

Solution:

Charge/ radius ratio of Be and Al is same because of diagonal relationship. Remaining statements are correct.

Question110

The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024 and 32824kJ mol $^{-1}$. The number of valence electrons in the clement is: [Sep. 03, 2020 (II)]

Options:

A. 5

B. 4

C. 3

D. 2

Answer: C

Solution:

As difference in 3^{id} and 4^{th} ionisation energies is high so atom contains 3 valence electrons.

Question111

In general, the property (magnitudes only) that shows an opposite trend in comparison to other properties across a period is:

[Sep .02,2020(1)]

Options:

A. lonization enthalpy

B. Electronegativity

C. Electron gain enthalpy

D. Atomic radius

Answer: D

Solution:

On moving left to right along a period in the periodic table atomic radius decreases while electronegativity, electron gain enthalpy and ionisation enthalpy increases, along a period.

Question112

Three elements X , Y and Z are in the 3^{nd} period of the periodic table. The oxides of X , Y and Z , respectively, arc basic, amphoteric and acidic. The correct order of the atomic numbers of X, Yand Zis :
[Sep. 02,2020(II)]

Options:

A. $Z < Y < X$

B. $X < Y < Z$

C. $X < Z < Y$

D. $Y < X < Z$

Answer: B

Solution:

On moving left to right in a period, the acidic character of oxides increases. 3^{nd} period element oxides.
Acidic character ∝ Atomic No.
So, X have minimum atomic number while Z have maximum atomic number.
Thus, the correct order of the atomic number is
$X < Y < Z$

Question113

The element with $Z = 120$ (not yet discovered) will be an\/a:
[Jan. 12,2019 (I)]

Options:

A. Inner-transition metal

B. Alkalineearth metal

C. Alkalimetal

D. Transition metal

Answer: B

Solution:

Elements with $Z = 120$ will belong to alkaline earth metals.
Its electronic con figuration may be represented as [Og] $8s^2$

Question114

The correct order of the atomic radii of C, Cs, Al , and S is:
[Jan. 11, 2019 (I)]

Options:

A. $C < S < Al < Cs$

B. $S < C < Cs < Al$

C. $S < C < Al < Cs$

D. $C < S < Cs < Al$

Answer: A

Solution:

On going down the group, size increases while going from left to right in a period size decreases, so order is
$C < S < Al < CCs$

Question115

The correct option with respect to the Pauling electronegativity values of the elements is:
[Jan. 11,2019 (II)]

Options:

A. $Te > Se$

B. $Ga < Ge$

C. $Si < Al$

D. $P > S$

Answer: B

Solution:

Correct order of electronegativity values of the elements is
$Si > Al$; $S > P$; $Se > Te$; $Ge > Ga$

Question116

The 71^{st} electron of an element X with an atomic number of 71 enters into the orbital:
[Jan.10,2019 (II)]

Options:

A. 6p

B. 4f

C. 5d

D. 65

Answer: C

Solution:

$_{71}X = [Xe]6s^2 4f^{14} 5d^1$
∴ Orbital occupied by last e^- is $5d$.

Question117

In general, the properties that decrease and increase down a group in the periodic table, respectively, are:
[Jan. 9,2019(I)]

Options:

A. atomic radius and electronegativity.

B. electron gain enthalpy and electronegativity.

C. electronegativity and atomic radius.

D. electronegativity and electron gain enthalpy.

Answer: C

Solution:

Generally, electronegativity decreases down the group as the size increases. This can also be formulated as:
Electronegativity $\propto \frac{1}{\text{size}}$

Question118

When the first electron gain enthalpy ($\Delta_{eg}H$) of oxygen is $-141 \mathrm{kJ} / \mathrm{mol}$, its second electron gain enthalpy is:
[Jan. 09,2019(II)]

Options:

A. a more negative value than the first

B. almost the same as that of the first

C. negative, but less negative than the first

D. a positive value

Answer: D

Solution:

The second electron gain enthalpy of oxygen is positive as energy has to be added for the addition of another electron.

Question119

The group number, number of valence electrons, and valency of an element with atomic number 15 , respectively, are:
[April 12, 2019 (I)]

Options:

A. 16,5 and 2

B. 15,5 and 3

C. 16,6 and 3

D. 15,6 and 2

Answer: B

Solution:

Phosphorus has atomic number 15. Its group number is 15, number of valence electrons is 5 and valency is 3.

Question120

The IUPAC symbol for the element with atomic number 119 would be : [April 8, 2019 (II)]

Options:

A. uue

B. une

C. unh

D. uun

Answer: A

Solution:

Symbol for 1 is u and for 9 is e.
∴ IUPAC symbol for 119 is uue.

Question121

In comparison to boron, berylium has: [April 12, 2019 (II)]

Options:

A. lesser nuclear charge and lesser first ionisation enthalpy.

B. greater nuclear charge and lesser first ionisation enthalpy.

C. greater nulear charge and greater first ionisation enthalpy.

D. lesser nuclear charge and greater first ionisation enthalpy.

Answer: D

Solution:

Nuclear charge : $B > Be$
$Be = 1s^2 2s^2$ (more stable)
$B = 1s^2 2s^2 2p^1$
$\therefore$ Ionisation energy of Be is greater than B due to ns^2 outer electronic configuration.

Question122

The element having greatest difference between its first and second ionization energies, is:
[April 9,2019(I)]

Options:

A. Ca

B. Sc

C. Ba

D. K

Answer: D

Solution:

Alkali metals have high difference in first and second ionisation energy as they achieve stable noble gas configuration after first ionisation.

Question123

The correct order of electron affinity is:
[Online April 15, 2018 (II)]

Options:

A. $O > F > Cl$

B. $F > O > Cl$

C. $F > Cl > O$

D. $Cl > F > O$

Answer: D

Solution:

On moving from left to right across a period, the electron affinity becomes more negative. On moving from top to bottom in a group, the electron affinity becomes less negative. Chlorine has exceptionally more negative electron affinity than fluorine, because adding an electron to fluorine ($2p$ orbital) causes greater repulsion than adding an electron to chlorine ($3p$ orbital) which is larger in size.

Question124

For Na^{+}, Mg^{2+}, F^{-} and O^{2-}; the correct order of increasing Ionic radii is : [Online April 15, 2018(I)]

Options:

A. $O^{2-} < F^{-} < Na^{+} < Mg^{2+}$

B. $Na^{+} < Mg^{2+} < F^{-} < O^{2-}$

C. $Mg^{2+} < Na^{+} < F^{-} < O^{2}$

D. $Mg^{2+} < O^{2-} < Na^{+} < F$

Answer: C

Solution:

All species are isoelectronic (10e-).
In isoelectronic series, when negative charge increases the radius of ion increases.
$\therefore Mg^{2+} < Na^{+} < F^{-} < O^{2}$

Question125

Both lithium and magnesium display several similar properties due to the diagonal relationship; however, the one which is incorrect is : [2017]

Options:

A. Both form basic carbonates

B. Both form soluble bicarbonates

C. Both form nitrides

D. Nitrates of both Li and Mg yield NO_2 and O_2 on heating

Answer: A

Solution:

Mg can form basic carbonate
$3MgCO_3 \cdot Mg(OH)_2.3H_2O\downarrow$
While Li can form only carbonate (Li_2CO_3), not basic carbonate.

Question126

The group having isoelectronic species is : [2017]

Options:

A. $O^{2-}, F^-, Na^+, Mg^{2+}$

B. O, F^-, Na, Mg^+

C. O^{2-}, F^-, Na, Mg^{2+}

D. O^-, F^-, Na^+, Mg^{2+}

Answer: A

Solution:

Isoelectronic species have same no. of electrons.
Ions ⇒
O^{2-} - $8+2$
F^- - $9+1$
Na^+ - $11-1$
Mg^{2+} - $12-2$
No. of e^-⇒μ
10 - O^{2-},
10 - F^-,
10 - Na^+,
10 - Mg^{2+}
Thereforeare isoelectronic.

Question127

Consider the following ionization enthalpies of two elements 'A' and 'B'.

Element	Ionization	enthalpy	(kJ/mol)
	1 st	2 nd	3 rd
A	899	1757	14847
B	737	1450	7731

[Online April 8, 2017|

Options:

A. Both 'A' and 'B' belong to group- 1 where ' B ' comes below 'A:

B. Both 'A' and 'B' belong to group- 1 where 'A' comes below 'B'.

C. Both 'A' and 'B' belong to group- 2 where ' B ' comes below 'A?

D. Both 'A' and ' B ' belong to group- 2 where 'A' comes below 'B'.

Answer: C

Solution:

Generally, the ionization enthalpies or energy increases from left to right in a period and decreases from top to bottom in a group. Several factor such as atomic radius, nuclear charge, shielding effect are responsible for change of ionization enthalpies. Here, Ist ionization enthalpy of A and B is greater than group I (Li $520 \mathrm{kJ} \mathrm{~mol}^{-1}$ to Cs $374 \mathrm{kJ} \mathrm{~mol}^{-1}$), which means element A and B belong to group -2 and all three given ionization enthalpy values are less for element B means B will come below A.

Question128

The electronic configuration with the highest ionization enthalpy is: [Online April 9, 2017]

Options:

A. $[\mathrm{Ne}]\ 3 s^2 3 p^{\prime}$

B. $[\mathrm{Ne}]\ 3 s^2 3 p^2$

C. $[\mathrm{Ne}]\ 3 s^2 3 p^3$

D. $[\mathrm{Ar}] 3 d^{10} 4 s^2 4 p^3$

Answer: C

Solution:

The smaller the atomic size, larger is the value of IP. Further the atoms having half filled or fully filled orbitals are comparatively more stable, hence more energy is required to remove the electron from such atoms.

Question129

Which of the following atoms has the highest first ionization energy? [2016]

Options:

A. K

B. Sc

C. Rb

D. N a

Answer: B

Solution:

Alkali metals have the lowest ionization energy in each period, on the other hand Sc is a d - block element. Transition metals have smaller atomic radii and higher nuclear charge leading to high ionisation energy.

Question130

The following statements concern elements in the periodic table. Which of the following is true? [Online April 10,2016]

Options:

A. For Group 15 clements, the stability of +5 oxidation state increases down the group

B. Elements of Group 16 have lower ionization enthalpy values compared to those of Group 15 in the corresponding periods.

C. The Group 13 elements are all metals

D. All the elements in Group 17 are gases.

Answer: B

Solution:

Due to extra stable half-filled p -orbital electronic configurations of Group 15 elements, larger amount of energy is required to remove electrons compared to Group 16 elements.

Question131

The ionic radii (in Å) of N^{3-}, O^{2-} and F^{-} are respectively:
[2015]

Options:

A. 1.71, 1.40 and 1.36

B. 1.71, 1.36 and 1.40

C. 1.36, 1.40 and 1.71

D. 1.36, 1.71 and 1.40

Answer: A

Solution:

For isoelectronic species, size of anion increases as negative charge increases. Thus the correct order is (a).

Question132

In the long form of the periodic table, the valence shell electronic configuration of $5s^2 5p^4$ corresponds to the element present in:
[Online April 10, 2015]

Options:

A. Group 16 and period 6

B. Group 17 and period 6

C. Group 16 and period 5

D. Group 17 and period 5

Answer: C

Solution:

Tellurium (Te) has $5s^2 5p^4$ valence shell configuration. It belongs to group 16 and present in period 5 of the periodic table.

Question133

Similarity in chemical properties of the atoms of elements in a group of the periodic table is most closely related to:
[Online April 12, 2014]

Options:

A. atomic numbers

B. atomic masses

C. number of principal energy levels

D. number of valence electrons

Answer: A

Solution:

If elements are arranged in order of their increasing atomic numbers, element coming at intervals of 2, 8, 8, 18, 18, 32 and 32 will have similar physical and chemical

----

Question134

Which of the following series correctly represents relations between the elements from X to Y ?

$X \rightarrow Y$
[Online April 11,2014]

Options:

A. $_3Li \rightarrow {}_{19}K$ Ionization enthalpy increases

B. $_9F \rightarrow {}_{35}Br$ Electron gain enthalpy (negative sign) increases

C. $_6C \rightarrow {}_{32}Ge$ Atomic radii increases

D. 18Ar → 54X e Noble character increases

Answer: C

Solution:

On moving down in a group atomic radii increases.

Question135

Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species O^{2-}, S^{2-}, N^{3-}, P^{3-}?
[Online April 12, 2014]

Options:

A. $O^{2^2} < N^{3-} < S^{2-} < P^3$

B. $O^{2-} < P^{3-} < N^3 < S^2$

C. $N^{3-} < O^{2-} < P^{3-} < S^{2-}$

D. $N^{3-} < S^{2-} < O^{2-} < P^{3-}$

Answer: A

Solution:

For isoelectronic species ionic radii decreases as the charge on ion decreases. Further on moving down in a group ionic radii increases. Hence the correct order is O^2
$< N^{3-} < S^{2-} < P^3$

Question136

Which one of the following has largest ionic radius?
[Online April 19,2014]

Options:

A. Li^+

B. O_2^{2-}

C. B^{3+}

D. F^-

Answer: B

Solution:

The ionic radii of elements exhibit the same trend as the atomic radii. The atomic size generally decreases across a period for the elements of the second period. It is because, within the period, the outer electrons are in the same valence shell and the effective nuclear charge increases as the atomic number increases resulting in the increased attraction of electrons to the nucleus. The size of an anion will be larger than that of the parent atom because the addition of one or more electrons would result in increased repulsion among the electrons and a decrease in effective nuclear charge.
In given question option A and C are cations in the second period. So they will have less ionic radius than that of the

anions o same period. oxygen falls left to the Fluorine and also has one extra negative charge (O_2^{-2}) than Fluorine ion (F^-)

Question137

Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar ?
[2013]

Options:

A. $Ca < S < Ba < Se < Ar$

B. $SSe < Ca < Ba < Ar$

C. $Ba < Ca < Se < S < Ar$

D. $Ca < Ba < S < Se < Ar$

Answer: C

Solution:

On moving down a group size increases, hence ionisation enthalpy decreases. Hence $Se < S$ and $Ba < Ca$.
Further, Ar being an inert gas has maximum IE.

Question138

The order of increasing sizes of atomic radii among the elements O, S, Se and As is:
[Online April 22, 2013]

Options:

A. $As < S < O < Se$

B. $Se < S < As < O$

C. $O < S < As < Se$

D. $O < S < Se < As$

Answer: D

Solution:

On moving down in a group atomic radii increases due to successive addition of extra shell hence, order is
$O < S < Se$

Further, As is in group 15 having one less electron in its p orbital compared to group 16 , hence as has higher atomic radii than group 16 elements.
i.e., $O < S < Se < As$

Question139

Which is the correct order of second ionization potential of C, N , O and F in the following?
[Online May 26, 2012; April 23, 2013]

Options:

A. $O > N > F > C$

B. $O > F > N > C$

C. $F > O > N > C$

D. $C > N \geq O > F$

Answer: B

Solution:

The second ionization potential means removal of electron from cation
$C^{+} = 1s^2 2s^2 2p^1$; $N^{+} = 1s^2 2s^2 2p^2$
$O^{+} = 1s^2 2s^2 2p^3$; $F^{+} = 1s^2 2s^2 2p^4$
Therefore, the correct order is
$O > F > N > C$

Question140

The electron affinity of chlorine is 3.7eV .1 gram of chlorine is completely converted to Cl^{-} ion in a gaseous state. ($1eV = 23.06 kcal\ mol^{-1}$) Energy released in the process is
[Online May 7,2012]

Options:

A. 4.8kcal

B. 7.2kcal

C. 8.2k cal

D. 2.4kcal

Answer: D

Solution:

Number of moles of chlorine $= \frac{1}{35.5}$ mol

Given, $1\mathrm{eV} = 23.06\mathrm{kcal\ mol}^{-1}$

$3.7\mathrm{eV} = 3.7 \times 23.06\mathrm{kcal\ mol}^{-1}$

i.e. 1 mole of chlorine releases energy

$= 3.7 \times 23.06\mathrm{kcal}$

$\therefore \frac{1}{35.5}$ mole of chlorine will release energy

$= \frac{1}{35.5} \times 3.7 \times 23.06\mathrm{kcal} = 2.4\mathrm{kcal}$

Question141

Which among the following elements has the highest first ionization enthalpy? [Online May 12, 2012]

Options:

A. Nitrogen

B. Boron

C. Carbon

D. Oxygen

Answer: A

Solution:

Due to stable $2s^2 2p^3$ configuration (half filled p -orbital). Nitrogen atom has highest energy.

Question142

The correct order of electron gain enthalpy with negative sign of F, Cl, Br and I, having atomic numbers 9, 17, 35 and 53 respectively, is: [2011RS]

Options:

A. $F > Cl > Br > I$

B. $Cl > F > Br > I$

C. $Br > Cl > I > F$

D. $I > Br > Cl > F$

Answer: B

Solution:

As we move down in a group, electron gain enthalpy becomes less negative because the size of the atom increases and the distance of added electron from the nucleus increases. Negative electron gain enthalpy of F is less than Cl. This is due to the fact that when an electron is added to F, the added electron goes to the smaller $n = 2$ energy level and experiences significant repulsion from the other electrons present in this level. In Cl, the electron goes to the larger $n = 3$ energy level and consequently occupies a larger region of space leading to much less electron-electron repulsion. So the correct order is
$Cl > F > Br > I$

Question143

The correct sequence which shows decreasing order of the ionic radii of the elements is
[2010]

Options:

A. $Al^{3+} > Mg^{2+} > Na^{+} > F^{-} > O^{2-}$

B. $Na^{+} > Mg^{2+} > Al^{3+} > O^{2-} > F$

C. $Na^{+} > F^{-} > Mg^{2+} > O^{2-} > Al^{3+}$

D. $O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}$

Answer: D

Solution:

All the given species contains $10e^{-}$ each i.e. isoelectronic. For isoelectronic species anion having high negative charge is largest in size and the cation having high positive charge is smallest.

Question144

In which of the following arrangements, the sequence is not strictly according to the property written against it?
[2008, Online May 7,2012]

Options:

A. $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power

B. $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

C. $HF < HCl < HBr < HI$: increasing acid strength

D. B < C < O < N : increasing first ionisation enthalpy.

Answer: B

Solution:

Correct order of increasing basic strength is
$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

Question145

Following statements regarding the periodic trends of chemical reactivityofalkali metals and halogens are given. Which of these statements gives the correct picture?
[2006]

Options:

A. Chemical reactivity increases with increase in atomic number down the group in both the alkali metals and halogens

B. In alkali metals the reactivity increases but in the halogens it decreases with increase in atomic number down the group

C. The reactivity decreases in the alkali metals but increases in the halogens with increase in atomic number down the group

D. In both, alkalimetals and halogens chemical reactivity decreases with increase in atomicnumber down the group

Answer: B

Solution:

The alkali metals are highlyreactive because their first ionisation potential is very low and hence they have great tendency to give up electron to form unipositive ion. Note: On moving down in group I from Li to C s, ionisation enthalpy decreases hence the reactivity increases. The halogens are most reactive elements due to their low bond dissociation energy, high electron affinity and high enthalpy of hydration of halide ion. However, their reactivity decreases with increase in atomic number.

Question146

The increasing order of the first ionization enthalpies of the elements B, P, S and F (Lowest first) is
[2006]

Options:

A. $B \le P < S \le F$

B. $B \le S < P < F$

C. $F < S \le P < B$

D. $P < S < B < F$

Answer: B

Solution:

The correct order of ionisation enthalpies is
$F > P > S > B$
Note: On moving along a period ionization enthalapy increases from left to right and decreases from top to bottom in a group. But this trend breaks up in case of such element having fully or half filled stable orbitals.
In this case, P has a stable half filled electronic configuration, hence its ionisation enthalapy is greater in comparision to S. Therefore the correct order is
$B < S < P < F$

Question147

Which of the following oxides is amphoteric in character? [2005]

Options:

A. SnO_2

B. SiO_2

C. CO_2

D. CaO

Answer: A

Solution:

CaO is basic as it formsstrong base $Ca(OH)_2$ on reaction with water.
$CaO + H_2O \longrightarrow Ca(OH)_2$
CO_2 is acidic as it dissolves in water forming unstable carbonic acid.
$H_2O + CO_2 \longrightarrow H_2CO_3$
Silica (SiO_2) is insoluble in water and acts as a very weak acid.
SnO_2 is amphoteric as it reacts with both acid and base.
$SnO_2 + 2H_2SO_4 \longrightarrow Sn(SO_4)_2 + 2H_2O$
$SnO_2 + 2KOH \longrightarrow K_2SnO_3 + H_2O$

Question148

In which of the following arrangements, the order is NOT according to the property indicated against it?
[2005]

Options:

A. $L < Na < K < Rb$:
Increasing metallic radius

B. $I < Br < F < Cl$:
Increasing electron gain enthalpy (with negative sign)

C. $B < C < N < 0$
Increasing first ionization enthalpy

D. $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$
Increasing ionic size

Answer: C

Solution:

In a period the value of ionisation potential increases from left to right with breaks where the atoms have somewhat stable configuration. In the given list Nhas halffilled stable orbitals. Hence N has highest ionisation energy. Thus the correct order is
$B < C < O < N$
and not as given in option (c)

Question149

Which one of the following ions has the highest value of ionic radius?
[2004]

Options:

A. O^{2}

B. B^{3+}

C. Li^{+}

D. F^{-}

Answer: A

Solution:

O^{2-} and F^- are isoelectronic $(10c^-)$. Hence have same number of electron but different number of proton, F^- has 9 proton whercas 0^{2-} has 8 proton. Therefore F^- has greater nuclear charge compared to O^{2-}. So O^{2-} will have greater ionic radius, i.e. $O^{2-} > F^-$
Further Li^+ and B^{3+} are also isoelectronic $(2e^-)$ therefore, no. of proton in Li is 3 and in B is 5 , So,
$Li^+ > B^{3+}$
Hence, the correct order of atomic size is
$O^{2-} > F^{-} > Li^+ > B^{3+}$
Therefore, O^{2-} has the highest value of ionic radius.

Question150

Among Al_2O_3, SiO_2, P_2O_3 and SO_2 the correct order of acid strength is [2004]

Options:

A. $Al_2O_3 < SiO_2 < SO_2 < P_2O_3$

B. $SiO_2 < SO_2 < Al_2O_3 < P_2O_3$

C. $SO_2 < P_2O_3 < SiO_2 < Al_2O_3$

D. $Al_2O_3 < SiO_2 < P_2O_3 < SO_2$

Answer: D

Solution:

As the size decreases the basic nature of oxides changes to acidic nature i.c., acidic nature increases.
$\underbrace{SO_2 > P_2O_3}_{\text{Acidic}} > \underbrace{SiO_2}_{\text{Weak acidic}} > \underbrace{Al_2O_3}_{\text{Ampboteric}}$
SO_2 and P_2O_3 are acidic as their corresponding acids H_2SO_3 and H_3PO_3 are strong acids.

Question151

The formation of the oxide ion $0^{2-}(g)$ requires first an exothermic and then an endothermic step as shown below
$O(g) + e^- = O^-(g)\ \Delta H^0 = -142 kJ\ mol^{-1}$

$O^-(g) + e^- = O^{2-}(g)\ \Delta H^0 = 844 kJ\ mol^{-1}$
This is because
[2004]

Options:

A. O - ion will tend toresist the addition of another electron

B. Oxygen has high electron affinity

C. Oxygen is more elecronegative

D. O -ion has comparativelylarger size than oxygen atom

Answer: A

Solution:

O ion exerts a force of repulsion on the incoming electron. The energy is required to overcome it.

Question152

According to the periodic law of elements, the variation in properties of elements is related to their
[2003]

Options:

A. nuclear masses

B. atomic numbers

C. nuclear neutron-proton number ratios

D. atomic masses

Answer: B

Solution:

According to modern periodic law, the properties of the elements are repeated after certain regular intervals when these elements are arranged in order of their increasing atomic numbers.

Question153

Which one of the following is an amphoteric oxide?
[2003]

Options:

A. Na_2O

B. SO_2

C. B_2O_3

D. ZnO

Answer: D

Solution:

Na_2O (basic), SO_2 and B_2O_3 (acidic) and ZnO is amphoteric.
